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Optical dielectric planar waveguide process

Abstract

A method for depositing silicon oxynitride film structures is provided that is used to form planar waveguides. These film structures are deposited on substrates and the combination of the substrate and the planar waveguide is used in the formation of optical interposers and subassemblies. The silicon oxynitride film structures are deposited using low thermal budget processes and hydrogen-free oxygen and hydrogen-free nitrogen precursors to produce planar waveguides that exhibit low losses for optical signals transmitted through the waveguide of 1 dB/cm or less. The silicon oxynitride film structures and substrate exhibit low stress levels of less than 20 MPa.

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Background/Summary

(1) The present application is continuation of U.S. Utility patent application Ser. No. 18/092,492, filed on Jan. 3, 2023, entitled: “OPTICAL DIELECTRIC PLANAR WAVEGUIDE PROCESS” (OPE105C), which is continuation of U.S. Utility patent application Ser. No. 17/509,786, filed on Oct. 25, 2021, entitled: “OPTICAL DIELECTRIC PLANAR WAVEGUIDE PROCESS” (OPE105B), which is continuation of U.S. Utility patent application Ser. No. 16/932,970, filed on Jul. 20, 2020, entitled: “OPTICAL DIELECTRIC PLANAR WAVEGUIDE PROCESS” (OPE105A), which is continuation of U.S. Utility patent application Ser. No. 16/258,292, filed on Jan. 25, 2019, entitled: “OPTICAL DIELECTRIC PLANAR WAVEGUIDE PROCESS” (OPE105) and continuation in-part of and claims priority from U.S. Provisional Patent Application Ser. No. 62/621,659, filed on Jan. 25, 2018, entitled: “Optical Dielectric Interposer” (OPE010-PRO), U.S. Utility patent application Ser. No. 16/036,151, filed on Jul. 16, 2018, entitled: “Optical Dielectric Waveguide Structures” (OPE101), U.S. Utility patent application Ser. No. 16/036,179, filed on Jul. 16, 2018, entitled: “Methods for Optical Dielectric Waveguide Structures” (OPE102), U.S. Utility patent application Ser. No. 16/036,208, filed on Jul. 16, 2018, entitled: “OPTICAL DIELECTRIC WAVEGUIDE SUBASSEMBLY STRUCTURES” (OPE103), U.S. Utility patent application Ser. No. 16/036,234, filed on Jul. 16, 2018, entitled: “METHODS FOR OPTICAL DIELECTRIC WAVEGUIDE SUBASSEMBLY STRUCTURE” (OPE104), all of which are incorporated herein by reference.

FIELD OF THE INVENTION

(1) The present invention relates to optoelectronic communication systems, and more particularly to a planar waveguide structure that is used in submount assemblies in these systems. Optical dielectric interposers are formed from the integration and patterning of this planar waveguide structure with a substrate to form compact interposers and optical submount assemblies that provide low loss in optoelectronic packages used for optical signal routing and transmission.

BACKGROUND

(2) Waveguides are used in optical communication networks for the transmission and routing of optical signals. For the transmission of the optical signals over long distances, waveguides can take the form of optical fibers, thin strands of glass that are used to transfer data over distances that can span tens of kilometers. Within the networks of long range optical fibers are signal processing nodes that contain packaged photonic and optoelectronic circuits that are used to perform various functions such as to encode, send, receive, decode, multiplex, and de-multiplex, among other optical and electrical signal processing functions, the optical signals that are delivered to these processing nodes via the optical fibers. And within the optoelectronic circuits in these processing nodes, optical signals are transmitted via free space and through short lengths of waveguide. These short lengths of waveguide are used to guide signals to a variety of small packaged devices or components that can transfer, combine, split, and route optical signals as the demands of the network require.

(3) Optoelectronic packages at signal processing nodes in optical communications networks generally include an optical submount assembly, which typically consists of one or more optical die (such as lasers and photodetectors), and that can include either the means for the free space transmission of optical signals or the planar waveguides and associated optical routing components, all of which are enclosed in an hermetically-sealed cavity formed by a cap and a substrate. A submount assembly can include, for example, a substrate or interposer, the optical routing components, and the signal-generating and signal-receiving devices and components.

(4) Routing of optical signals from the optical fibers to components on the submount assembly have historically been accomplished via transmission in free space, and to some extent, via planar optical waveguides on the submount assembly. Optical transmission in free space can require lenses to focus and direct the optical signals between components in the optical circuits and can require large spatial volumes to accommodate these lenses. The large spatial volumes can lead to undesirably large package sizes for these optical circuits. Additionally, the transmission of the signals in free space can result in significant signal losses from uncontrolled scattering and reflection.

(5) Currently, the capability for fabricating planar waveguide structures of sufficient thickness with low stress is limited, and therefore, a need exists in the art of optoelectronic packaging for a planar waveguide structure that can be deposited onto a substrate, and from which compact and economical interposers and submount assemblies can be formed. Thus, there is a need in the art for a method to produce compact, planar optical waveguide structures that exhibit low optical loss without deformation of the underlying substrate.

Description

DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings illustrate various embodiments of systems, methods, and other aspects of the invention. It will be apparent to a person skilled in the art that the illustrated element boundaries (e.g., boxes, groups of boxes, or other shapes) in the figures represent one example of the boundaries. In some examples, one element may be designed as multiple elements, or multiple elements may be designed as one element. In some examples, an element shown as an internal

component of one element may be implemented as an external component in another, and vice versa.

(2) Various embodiments of the present invention are illustrated by way of example, and not limited by the appended figures, in which like references indicate similar elements, and in which:

(3) FIG. 1A-1C show cross-sectional schematic views of the inventive dielectric film structure for the formation of integrated planar waveguide structures: FIG. 1A a cross-sectional view of a single or multilayer dielectric top spacer layer structure for the inventive planar dielectric waveguide structure; FIG. 1B a cross-sectional view of a multilayer, repeating silicon oxynitride film structure for the inventive planar dielectric waveguide structure; and FIG. 1C a cross-sectional view of a single or multilayer dielectric bottom spacer layer structure for the inventive planar dielectric waveguide structure;

(4) FIG. 2A-2B show measured film stress in accordance with embodiments for FIG. 2A dielectric films deposited at various film thicknesses, and FIG. 2B dielectric films of various refractive indexes;

(5) FIG. 3A-3B show measured optical losses in accordance with embodiments for FIG. 3A dielectric films of various refractive indexes and FIG. 3B dielectric waveguide film structures with various bottom buffer layer film thicknesses;

(6) FIG. 4A-4C show steps for forming some embodiments of the inventive dielectric film structure FIG. 4A at low temperature and having low stress and low optical loss, FIG. 4B with each dielectric film deposited at low temperature and having low stress and low optical loss, and FIG. 4C that include a substrate with a buffer layer, one or more optional bottom spacer layers, a repeating stack of one or more dielectric layers, one or more optional top spacer layers, and an optional top layer, followed by patterning of the stack to form a waveguide;

(7) FIG. 5 shows a schematic of a typical plasma enhanced chemical vapor deposition system;

(8) FIG. 6 shows the measured refractive index for waveguide structures that were deposited over a range of the silicon-containing precursor gas flow;

(9) FIG. 7A-7B show cross sectional schematic views of embodiments of an integrated planar waveguide on a substrate: FIG. 7A without optical/electrical devices, and FIG. 7B with optical/electrical devices;

(10) FIG. 8A-8B show cross sectional schematic views of embodiments of integrated planar waveguides on a substrate with an interconnect layer in accordance with the inventive process: FIG. 8A without optical/electrical devices and FIG. 8B with optical/electrical devices;

(11) FIG. 9A-9B show cross sectional schematic views of embodiments of integrated planar waveguides on a substrate with interconnect layer and integrated electrical devices in the substrate in accordance with the inventive process: FIG. 9A without surface mounted optical or electrical devices and FIG. 9B with surface mounted optical or electrical device;

(12) FIG. 10A-10B show cross sectional schematic views of embodiments of integrated planar waveguides on a substrate with interconnect layer and integrated electrical devices in the substrate in accordance with the inventive process shown with interconnections between top surface mounted device and integrated electrical devices in the substrate: FIG. 10A shown without the top mounted optical or electrical devices in place, and FIG. 10B with top mounted optical or electrical device; also shown is the position of an optical fiber relative to the planar waveguide in an embodiment;

(13) FIG. 11A-11D show cross sectional schematic views of embodiments of a substrate with interconnect layer: FIG. 11A with inventive dielectric stack mounted via bond pads to the substrate as a discrete optical waveguide component, FIG. 11B with inventive dielectric stack mounted to the substrate as a discrete optical waveguide component and aligned with discrete optical and electrical devices, and aligned to an optical fiber, FIG. 11C with inventive dielectric stack mounted to the substrate as a discrete optical waveguide component, for which the substrate contains integrated electrical devices, and FIG. 11D with inventive dielectric stack mounted to the substrate as a

discrete optical waveguide component and aligned with discrete optical and electrical devices, and aligned to an optical fiber for an embodiment in which the substrate contains integrated electrical devices;

(14) FIG. 12A-12B show steps in the fabrication of embodiments of providing a patterned dielectric waveguide structure FIG. 12A on a substrate with one or more integrated devices in the substrate that are coupled to an interconnect layer, and FIG. 12B on a substrate with one or more integrated devices in the substrate that are coupled to the inventive planar waveguide through an interconnect layer and a device, and to an optical fiber that is configured to interface with the planar waveguide;

(15) FIG. 13A shows a perspective schematic view of a substrate with patterned inventive dielectric waveguide structure, with a v-groove for mounting and alignment of an optical fiber and with mechanical stops for the mounting and alignment of optical and electrical devices and die, and FIG. 13B shows a cross sectional schematic view of embodiments of integrated planar waveguide structures on a substrate with alignment mark and stops for alignment of optical/electrical devices;

(16) FIG. 14 shows steps in the fabrication of embodiments of the inventive dielectric film structures for providing patterned dielectric waveguides on substrates with features for the alignment of optical and electrical devices;

(17) FIG. 15A-15B show cross sectional schematic views of embodiments of integrated planar waveguides on a substrate with integrated heat sink layer FIG. 15A on the substrate, and FIG. 15B within the interconnect layer;

(18) FIGS. 16A-16B show steps in the fabrication of embodiments of the inventive dielectric film structure for providing patterned dielectric waveguide structures with FIG. 16A interconnection layer formed on a thermal conductive layer, and FIG. 16B a high thermal conductivity dielectric layer within the interconnect layer;

(19) FIG. 17A-17D show a cross sectional schematic view of embodiments of integrated planar waveguides on a substrate shown FIG. 17A with unpatterned dielectric waveguide stack, FIG. 17B patterned dielectric waveguide structure with resulting cavity shown in cross section and, in the inset, in a perspective view, FIG. 17C with patterned dielectric waveguide structure and with mounted optical/electrical die within the cavity, and FIG. 17D with patterned dielectric waveguide structure, mounted optical/electrical die, and hermetic sealing cap;

(20) FIG. 18 shows steps in the fabrication of embodiments of the inventive dielectric film structure for the formation of integrated planar waveguides and mechanical support structures to support hermetic sealing.

SUMMARY

(21) Embodiments of the present invention are directed to the fabrication of integrated planar dielectric waveguides that are formed and patterned on substrates. The combination of an integrated planar waveguide and a substrate, to form an optical dielectric interposer, serves as a subcomponent of an optical submount assembly for an optoelectronic package. Optical losses in dielectric waveguides are minimized with the use of nitrogen and oxygen precursors that do not contain hydrogen.

(22) The present invention is based, in part, on the development of a dielectric waveguide structure that transmits optical signals with low loss, is integrated into a substrate and thereby reduces fabrication costs, is deposited at low processing temperatures of less than 400° C., and preferably less than or at approximately 300° C., and is fabricated with low stress to prevent stress-induced delamination of the film structure and deformation of the substrate. The inventive structure and method for producing the structure enables the fabrication of waveguides over a wide range of thicknesses up to 20 micrometers and greater. The benefits of this capability, as further described herein, provide superior optical and mechanical performance and superior economic benefits in comparison to the current state of the art.

(23) Waveguides that are fabricated using the inventive techniques described herein, have low

residual stresses and thus very thick layered structures can be deposited. The thicker, multilayer waveguide structures obtainable with the inventive method, coupled with low residual hydrogen concentrations in the deposited layers, allow for low optical loss. Further, the thick waveguides obtainable with the inventive method are provided at a thermal budget and can thus be fabricated on substrates that contain metallization layers and integrated electrical and optoelectrical devices.

DETAILED DESCRIPTION

(24) The ensuing description provides exemplary embodiment(s) only, and is not intended to limit the scope, applicability or configuration of the disclosure. Rather, the ensuing description of the exemplary embodiment(s) will provide those skilled in the art with an enabling description for implementing an exemplary embodiment. It being understood that various changes may be made in the function and arrangement of elements without departing from the spirit and scope as set forth in the appended claims.

(25) An “interposer” as used herein and throughout this disclosure refers to, but is not limited to, a substrate that provides mechanical support and electrical or optical interface routing from one or more electrical, optical, and optoelectrical devices to another. Interposers are typically used to route optical or electrical connections from various devices or die that are mounted on, or connected to, the interposer. An “optical interposer” is an interposer that provides for the optical interfacing between optical devices mounted or connected thereon.

(26) A “submount assembly” as used herein and throughout this disclosure refers to, but is not limited to, an assembly that includes a substrate, typically an interposer, that is populated with one or more optical, optoelectrical, and electrical devices.

(27) A “substrate” as used herein and throughout this disclosure refers to, but is not limited to, a mechanical support upon which an interposer is formed. Substrates may include, but not be limited to, silicon, indium phosphide, gallium arsenide, silicon, silicon oxide-on-silicon, silicon dioxide-on-silicon, silica-on-polymer, glass, a metal, a ceramic, a polymer, or a combination thereof. Substrates may include a semiconductor or other substrate material, and one or more layers of materials such as those used in the formation of an interconnect layer.

(28) An “optical die” as used herein and throughout this disclosure refers to, but is not limited to, a discrete optical device such as a laser or photodetector that can be positioned into a submount assembly as a component of an optical or optoelectronic circuit.

(29) An “optoelectronic package” as used herein and throughout this disclosure refers to, but is not limited to, an assembly that is typically hermetically sealed, and that typically includes a submount assembly and a cap; the package typically provides electrical, optical, or both electrical and optical interconnects for combining with external optoelectronic, electronic, and optical components as in, for example, an optical communications network, an optical circuit, or an electrical circuit.

(30) An “optical waveguide” as used herein and throughout this disclosure refers to, but is not limited to, a medium for transmitting optical signals.

(31) “Optical signals” as used herein and throughout this disclosure refers to, but is not limited to, electromagnetic signals typically in the infrared and visible light ranges of the electromagnetic spectrum that are encoded with information.

(32) A “semiconductor” as used herein and throughout this disclosure refers to, but is not limited to, a material having an electrical conductivity value falling between that of a conductor and an insulator. The material may be an elemental material or a compound material. A semiconductor may include, but not be limited to, an element, a binary alloy, a tertiary alloy, and a quaternary alloy. Structures formed using a semiconductor or semiconductors may include a single semiconductor material, two or more semiconductor materials, a semiconductor alloy of a single composition, a semiconductor alloy of two or more discrete compositions, and a semiconductor alloy graded from a first semiconductor alloy to a second semiconductor alloy. A semiconductor may be one of undoped (intrinsic), p-type doped, n-type doped, graded in doping from a first doping level of one type to a second doping level of the same type, and graded in doping from a

first doping level of one type to a second doping level of a different type. Semiconductors may include, but are not limited to III-V semiconductors, such as those between aluminum (Al), gallium (Ga), and indium (In) with nitrogen (N), phosphorous (P), arsenic (As) and tin (Sb), including for example GaN, GaP, GaAs, InP, InAs, AlN and AlAs.

(33) “Silicon oxynitride” as used herein and throughout this disclosure refers to, but is not limited to, a dielectric material that is formed by a combination of constituent elements of silicon, oxygen, and nitrogen. In some instances, the term “silicon oxynitride” can refer to silicon oxides and silicon nitrides in the general sense that silicon oxides and silicon nitrides are silicon oxynitrides with very low or insignificant levels of either the nitrogen in the case of silicon oxides, and oxygen in the case of silicon nitrides. Film properties, such as the refractive index, can be controlled or varied by varying the concentrations and the ratios of the constituent elements of silicon, oxygen, and nitrogen, and to some extent, by the concentrations of impurities in the films. The removal of nitrogen or the reduction of nitrogen to low levels, for example, in one film of a film stack, does not change the designation of the material as silicon oxynitride within the context of this disclosure. Similarly, the removal of oxygen or the reduction of oxygen to very low levels does not change the designation of the resulting material as a silicon oxynitride. Materials with low or unmeasurable levels of either nitrogen or oxygen should, therefore, be viewed as silicon oxynitrides within the context of this disclosure. The ratio of silicon to oxygen to nitrogen in silicon oxynitride films can vary over a wide range and variations in the ratio of these constituent elements can lead to variations in the refractive indices of silicon oxynitride films as described herein. The concentrations of impurities in the films, from the deposition processes used to form the films, can also influence the indices of refraction of the silicon oxynitride films. Silicon oxynitride is electrically insulating and optically transparent.

(34) “Silicon oxide” as used herein and throughout this disclosure refers to, but is not limited to, a dielectric material that is formed from a combination of silicon and oxygen, and in some instances may contain other elements such as hydrogen, for example, as a byproduct of the deposition method. In its most common form, the ratio of oxygen to silicon is 2:1 (silicon dioxide) but variations in this ratio remain within the scope of the definition of silicon oxide as used for the silicon oxide films in this disclosure. Similarly, variations in stoichiometry are to be anticipated and applicable for films specifically referred to in this disclosure as silicon dioxide.

(35) “Silicon nitride” as used herein and throughout this disclosure refers to, but is not limited to, a dielectric material that is formed from a combination of silicon and nitrogen, and in some instances may contain other elements such as hydrogen, for example, as a byproduct of the deposition method. In its most common form, the ratio of nitrogen to silicon is 4:3, but variations in this ratio remain within the scope of the definition of silicon nitride as used for the silicon nitride films in this disclosure.

(36) A “metal” as used herein and throughout this disclosure refers to, but is not limited to, a material (element, compound, and alloy) that has good electrical and thermal conductivity. This may include, but not be limited to, gold, chromium, aluminum, silver, platinum, nickel, copper, rhodium, palladium, tungsten, and combinations of such materials.

(37) An “electrode”, “contact”, “track”, “trace”, or “terminal” as used herein and throughout this disclosure refers to, but is not limited to, a material having good electrical conductivity and that is typically, optically opaque. This includes structures formed from thin films, thick films, and plated films for example of materials including, but not limited to, metals such as gold, chromium, aluminum, silver, platinum, nickel, copper, rhodium, palladium, tungsten, and combinations of such materials. Other electrode configurations may employ combinations of metals, for example, a chromium adhesion layer and a gold electrode layer.

(38) A “precursor” as used herein and throughout this disclosure is a material in gaseous or vapor form that contains one or more of the constituent elements of a material targeted for deposition. Molecular silane (SiH_4), for example, contains elements of silicon and hydrogen, and is a

widely used precursor for depositing films in chemical vapor deposition processes, for example, that contain silicon (such as silicon, silicon dioxide, silicon nitride, silicon oxynitride, etc.) Deposition of films such as silicon dioxide, for example, requires at least a second precursor to provide the oxygen in applications in which silane is used since the silane does not contain oxygen. Molecular oxygen (O₂) is often used as a precursor with silane in the deposition of thin films of silicon dioxide to provide the oxygen atoms required for the silicon oxide formation. It is important to note that not all of the elements contained in the precursor or precursors are required or desired in the deposited films. In the case of silane, for example, when used as a precursor to deposit thin films of silicon dioxide, residual hydrogen can lead to a reduction in the quality of the films relative to stoichiometrically pure films without the hydrogen.

(39) References to “an embodiment”, “another embodiment”, “yet another embodiment”, “one example”, “another example”, “yet another example”, “for example” and so on, indicate that the embodiment(s) or example(s) so described may include a particular feature, structure, characteristic, property, element, or limitation, but that not every embodiment or example necessarily includes that particular feature, structure, characteristic, property, element or limitation. Furthermore, repeated use of the phrase “in an embodiment” does not necessarily refer to the same embodiment.

(40) Planar optical waveguides that can be fabricated on substrates, and that can further be utilized in the formation of interposers, and ultimately in submount assemblies, offer the potential for significant reduction in fabrication complexity and in optoelectronic package sizes over current methods. The integration and patterning of planar waveguide structures on substrates allow for the transmission and distribution of optical signals within the subassemblies, which can eliminate or reduce the requirements to mount discrete optical components to the subassembly. Integrated waveguide structures also allow for the formation of optical device structures, such as filters, gratings, and spot size converters, for example, directly onto the substrate of the interposer or submount assembly, the integration of which can lead to significant cost advantages. Further, the ability to pattern and integrate planar waveguide structures on substrates that contain metal interconnect layers enables the use of integrated optoelectrical devices with the planar waveguides and the planar waveguide device structures fabricated on the substrates. This combination of integrated optical components with electrical interconnect layers requires low thermal budget fabrication processes so as to not damage the interconnect layers present in the substrates or interposers upon which the planar waveguides are fabricated.

(41) In an embodiment, a method for depositing a planar dielectric layer of silicon oxynitride on a substrate is provided which includes maintaining the substrate at a predetermined temperature and exposing the substrate surface to a plasma enhanced chemical vapor deposition process containing a silicon-containing precursor gas that may or may not contain hydrogen, a nitrogen-containing precursor that does not include hydrogen, and an oxygen-containing precursor that does not contain hydrogen. Advantages of the inventive low thermal budget process for forming the inventive dielectric waveguide structure **140** include predetermined temperatures that are sufficiently low to allow for substrates with metallization layers that would otherwise not be implementable with higher thermal budget processes.

(42) In general, the thermal budget of a manufacturing process is the total amount of thermal energy that is transferred to the substrate during an operation that is performed at elevated temperature, and is proportional to the temperature and duration of each of the steps in the given high temperature process. Lower temperature processes can typically withstand much longer durations without deleteriously impacting the substrate or features of the substrate that were formed in prior processes. Additionally, lower temperature processes in general, and more specifically, lower thermal budget processes, allow for features to be incorporated into the substrate that would not otherwise be available in cases in which high thermal budget processes are required. In the present inventive process and method, low thermal budget processing provides advantages in the

formation of dielectric film structures that include 1) a reduction in the potential for deformation of the substrates, and 2) the capability to incorporate integrated electrical devices and metallized layers into the substrate prior to the fabrication of the planar waveguide structures. High thermal budget processes in which the substrates are subjected to high temperatures, even for short durations, can lead to warpage and other forms of deformation of the substrate. Additionally, integrated electrical devices and metallization layers might also be subject to damage by processes at higher thermal budgets or require more robust and potentially more expensive processing to reduce this potential for damage. It is therefore advantageous for deposition processes for planar waveguide structures to have sufficiently low thermal budgets so as to not result in deformation or warping of the substrate, and so as not to result in damage to integrated electrical devices or metallized layers that are formed on the substrate prior to the formation of the waveguides. Additionally, the use of low thermal budget processes allows for the use of more economically favorable materials and processing techniques that may not have the robustness to withstand higher thermal budget processes. In the case of material selection, for example, the use of low thermal budget processes can allow for a variety of polymeric materials to be utilized that are commonly implemented in semiconductor fabrication schemes that would otherwise not withstand exposure to high temperature, or high thermal budget processes. Similarly, other materials and combinations of materials can be utilized with low thermal budget fabrication schemes that are susceptible to damage upon exposure to higher thermal budget processes.

(43) In applications in which electrical devices or metallization layers are incorporated into the substrate, low thermal budget processes offer significantly greater levels of flexibility in processing over higher thermal budget processes. In integrated device structures, the metallization layers are typically the most susceptible to the introduction of damage from high thermal budget processes, although other aspects of the device might also be affected by excursions to high temperatures.

(44) A metallized layer structure in or on a substrate typically consists of one or more patterned metal layers that are insulated with one or more layers of dielectric material to electrically isolate the metal layers. For example, a metallization layer structure can consist of multiple layers of patterned aluminum, copper, or other metals, or a combination of metals that are encapsulated within a dielectric material such as silicon dioxide, silicon nitride, a polymer, or other insulating material that provides electrical isolation between traces formed by the metal layers. The metallized layers with the dielectric insulation layers can be fabricated using either subtractive processing such as metal deposition followed by an etching step to pattern the metal layers or additive processing, such as damascene and dual-damascene processing in which dielectric intermetal layers are patterned, etched, and then filled with metal to form the metal traces that are subsequently used to interconnect devices that will ultimately be positioned or fabricated on the substrate. In the case of aluminum and alloys of aluminum, subtractive processes are most commonly used in which an aluminum layer is deposited, patterned, and etched to form a patterned metal interconnect layer and then covered with a dielectric to electrically isolate and protect, among other important aspects, the traces created in the aluminum layer. Conversely, in the case of copper metallization, damascene processes are most commonly used. In a damascene process, a dielectric layer is deposited, patterned, and etched to form trenches that are subsequently filled with copper to form the metal interconnect traces. The copper-filled dielectric structures are then subjected to a planarization step to remove excess copper and then capped with a dielectric layer to isolate and protect the copper layer below. For both the subtractive and additive processes, the steps are repeated for applications in which multiple metal layers are implemented in the structure. In cases in which multiple layers of metal are required, allowance must be made to vertically interconnect the metal layers as required by the circuit design. This overview of forming metallization layer structures is a simplified overview but nonetheless highlights some of the key aspects of metallization layer formation that can be used to form the metallization layer on a substrate prior to the formation of the inventive planar dielectric waveguide structure. The formation of metallization layers using

either subtractive or additive processes are well known in the art of integrated circuit manufacturing and applicable to the formation of metallization layers on substrates used to form optoelectronic interposers and subassemblies.

(45) The ultimate temperature for a substrate with one or more metal layers, in the extreme case, is limited by the melting point temperature for the metal, although other deleterious effects are often observed at temperatures much lower than the melting temperature for the metal. For applications in which metallization layers utilize aluminum, the melting point for which is approximately 660° C., low thermal budget processes are particularly enabling due to the susceptibility of the aluminum to reflow at temperatures much lower than the melting point. The reduction in appearance of other effects in addition to metal reflow, such as void formation, diffusion, and alloying with adjacent materials, for example, among other potential damage mechanisms are also minimized using lower thermal budget processes.

(46) In an embodiment, a method for depositing a planar dielectric layer of silicon oxynitride on a substrate is provided which includes maintaining the substrate at a predetermined temperature of less than 500 C and exposing the substrate surface to a plasma enhanced chemical vapor deposition process containing a silicon-containing precursor gas that may or may not contain hydrogen, a nitrogen-containing precursor that does not include hydrogen, and an oxygen-containing precursor that does not contain hydrogen. In another embodiment, a method for depositing a planar dielectric layer of silicon oxynitride on a substrate is provided which includes maintaining the substrate at a predetermined temperature of less than 400 C and exposing the substrate surface to a plasma enhanced chemical vapor deposition process containing a silicon-containing precursor gas that may or may not contain hydrogen, a nitrogen-containing precursor that does not include hydrogen, and an oxygen-containing precursor that does not contain hydrogen. And in yet another embodiment, a method for depositing a planar dielectric layer of silicon oxynitride on a substrate is provided which includes maintaining the substrate at a predetermined temperature of less than or approximately equal to 300 C and exposing the substrate surface to a plasma enhanced chemical vapor deposition process containing a silicon-containing precursor gas that may or may not contain hydrogen, a nitrogen-containing precursor that does not include hydrogen, and an oxygen-containing precursor that does not contain hydrogen. In embodiments, low thermal budget film deposition processes, as described herein, are used to form low stress film structures as shown in FIG. 1 for dielectric film structure **140**.

(47) Thick dielectric films of greater than a few micrometers for use as waveguides can provide transmission and routing pathways with desirably low levels of optical signal attenuation. Optical absorption is minimized in the current invention with the use of hydrogen-free precursors for oxygen and nitrogen that produce low levels of nitrogen-hydrogen bonding in deposited films, and when combined with the low stress levels of these deposited films, enable thick film structures as described in this disclosure to be fabricated. A low concentration of residual hydrogen-nitrogen bonding in the as-deposited films with the use of the hydrogen-free precursors provides the desirably low levels of optical absorption with a low thermal budget deposition method, and the means for deposited relatively thick dielectric layers that are low in stress. Thus, the inventive method provides a means for producing thick dielectric films that have low stress, low optical absorption, and that are fabricated with low thermal budget processes.

(48) Optical and electrical devices that are fabricated in, and utilized with, dielectric interposer layers that are formed with a low thermal budget enable a wide range of substrate functionality and integration. Integrated dielectric waveguides in dielectric interposers, for example, that can be fabricated with low stress and with low optical loss, allow for increased levels of integration. Low stress levels allow for minimal substrate deformation particularly in thin substrates. Low optical loss allows for higher density components and smaller overall package sizes for submount assemblies. Additionally, dielectric interposers that can be deposited at temperatures that do not lead to degradation of temperature sensitive metallization layers allow for the fabrication of

waveguides on substrates that include metallization. Substrates that contain integrated metallization layers includes a wide range of optical device assemblies that require electrical contact layers to interconnect optoelectrical devices to one another within the submount assembly, to devices within the substrate, and to other assemblies connected to the submount assembly. The ability to fabricate integrated waveguides on metalized CMOS structures, for example, enables an expansive range of functionality and applications in integrated communications technology.

(49) In exemplary embodiments, planar dielectric film structures of multiple layers of silicon oxynitride are formed on a substrate and patterned into waveguides using hydrogen-free precursors for the nitrogen and oxygen components of the deposited silicon oxynitride. The inventive processes can produce patterned waveguides on substrates with optical losses that are typically less than 1 dB/cm and that exhibit post-deposition stress levels of less than 20 MPa. The inventive processes in embodiments use plasma enhanced chemical vapor deposition (PECVD) with process chemistries that do not utilize hydrogen-containing nitrogen and oxygen precursors. Residual hydrogen in silicon oxynitride films has been linked to high levels of loss for optical signals transmitted through waveguides fabricated from silicon oxynitride reportedly due to the formation of high levels of nitrogen-hydrogen bonds in the deposited films. Nitrogen-hydrogen bonds in deposited films are known to absorb optical signals at wavelengths commonly used in optical fiber communication networks. In the inventive process, the incorporation of the hydrogen, and the formation of the nitrogen-hydrogen bonds, is limited in the deposited films by the use of nitrogen and oxygen precursor gases that do not contain hydrogen. The inventive process and method, therefore, provide a preferred approach to forming low loss optical waveguide structures by combining a dielectric stack structure and a low thermal budget process that minimizes nitrogen-hydrogen bonds in the silicon oxynitride. The low thermal budget process utilizes non-hydrogen-containing nitrogen and oxygen precursors during the dielectric film deposition to provide waveguides that exhibit low optical signal loss.

(50) Although the use of hydrogen-free precursors in the inventive dielectric film deposition process for the formation of the inventive film structure can result in significantly lower film deposition rates relative to processes that utilize hydrogen-containing precursors, the combination of the inventive process with these film structures enable a range of innovations as a result of the reductions in optical signal loss and residual stresses in the deposited films. The use of hydrogen-free nitrogen precursors and hydrogen-free oxygen precursors greatly reduces the formation of nitrogen-hydrogen bonds in the deposited silicon oxynitride films, and therefore, significantly reduces the absorption of optical signals in waveguides fabricated from these films in comparison to processes that utilize hydrogen-containing precursors.

(51) Further areas of applicability of the present invention will become apparent from the detailed description provided herein. It should be understood that the detailed description of exemplary embodiments is intended for illustration purposes only and is, therefore, not intended to necessarily limit the scope of the present invention.

(52) An embodiment of the inventive dielectric waveguide structure is shown in FIG. 1. The inventive dielectric waveguide structure is a stack of dielectric films deposited on a substrate **110** to form optical dielectric interposer **100**. In an embodiment, the substrate is silicon. In other embodiments, the substrate is GaAs, InP, SiGe, SiC, or another semiconductor. In yet other embodiments, the substrate is aluminum nitride, aluminum oxide, silicon dioxide, quartz, glass, sapphire, or another ceramic or dielectric material. In yet other embodiments, the substrate is a metal. And in yet other embodiments, the substrate is a layered structure of one or more of a semiconductor, a ceramic, and a metal. It is to be understood that the substrate can be any material that provides a suitable mechanical support. It is to be further understood that a substrate with an interconnect layer that contains electrical lines and traces, separated with intermetal dielectric material, is a substrate.

(53) The optical dielectric interposer **100** includes a planar waveguide structure formed on

substrate **110**. In the preferred embodiment, the planar waveguide structure includes a buffer layer **130**, spacer layer **138**, a repeating stack of silicon oxynitride films **142**, a top spacer layer **150**, and an optional top layer **158**.

(54) In preferred embodiments, buffer layer **130** is one or more layers of silicon dioxide or silicon oxynitride. In some embodiments, the buffer layer is a layer of silicon oxynitride. In a preferred embodiment, the buffer layer **130** is a silicon oxynitride layer, 5000 nm in thickness, with an index of refraction of 1.55. In other embodiments, the buffer layer **130** is silicon oxynitride with refractive index of 1.55 and is thicker than 2000 nm. In other embodiments, the buffer layer **130** is a silicon dioxide layer with a refractive index of approximately 1.445. In other embodiments, the buffer layer **130** is a silicon dioxide layer with a refractive index of approximately 1.445 that is greater than 2000 nm in thickness. In a preferred embodiment, the buffer layer **130** is a silicon dioxide layer that is approximately 4000 nm in thickness and with a refractive index of approximately 1.445.

(55) Buffer layer **130** can be a composite layer of one or more layers of silicon dioxide or silicon oxynitride with varying thicknesses that in some embodiments sum to greater than 4000 nm in total thickness. Similarly, the buffer layer **130**, in some preferred embodiments, can be a composite layer of one or more layers with varying refractive index, that when combined, provide a total thickness of greater than 4000 nm and a composite refractive index in the range of 1.4 to 2.02.

(56) In preferred embodiments, spacer layer **138** is one or more layers of silicon dioxide or silicon oxynitride. In a preferred embodiment, the spacer layer **138** is a single spacer layer **138a** of silicon oxynitride, 500 nm in thickness, with an index of refraction of 1.55. In some embodiments, single spacer layer **138a** is a layer of a single material, such as silicon dioxide. In other preferred embodiments, single spacer layer **138a** is a layer of silicon oxynitride. In yet other preferred embodiments, the single spacer layer **138a** is a layer of silicon oxynitride with refractive index of 1.55 with thickness of 500 nm. In yet other embodiments, single spacer layer **138a** is a layer of silicon oxynitride with thickness in the range of 0 to 1000 nm. Although in preferred embodiments, a spacer layer **138** is included in the structure, in some other embodiments, the spacer layer **138**, can be combined with the buffer layer, can be made very thin, or is not included (FIG. **1C**).

(57) Spacer layer **138** can be a composite spacer layer **138b** of one or more layers of silicon oxynitride or silicon dioxide. In an embodiment, composite spacer layer **138b** includes two layers of silicon oxynitride with thicknesses of 250 nm and with a composite refractive index of approximately 1.55. In some embodiments, the sum of the thicknesses of the two layers in composite spacer layer **138b** is in the range of 1 to 1000 nm (FIG. **1C**).

(58) Similarly, the spacer layer **138** can be a composite layer **138c** of three or more layers with the same or varying thicknesses and refractive indices, that when combined, provide a total thickness in the range of 1 nm to 1000 nm and a composite refractive index in the range of 1.4 to 2.02 (FIG. **1C**).

(59) The combined thicknesses of the buffer layer **130** and the spacer layer **138** in embodiments provide spatial separation between the core repeating stack **142** and the substrate **110** and reduce, minimize, or eliminate the interaction of the transmitted optical signal with the substrate **110**. The transmission of optical signals with low optical loss through the repeating structure **142** requires some degree of confinement of the signal to the waveguide with minimal interaction of the optical signals with the substrate **110** in embodiments for which the optical signals are attenuated in the substrate material. Silicon and some other semiconductors, and metal layers in the interconnect layers, for example, can lead to significant attenuation of optical signals. The combined thicknesses of the buffer layer **130** and the spacer **138** provide spatial isolation between the substrate materials and the upper layers of the inventive dielectric stack structure to reduce the interaction of transmitted optical signals with materials in the substrate that can lead to attenuation.

(60) Dielectric stack **142** forms the core of the inventive waveguide structure through which optical signals can be transmitted with low optical loss. In preferred embodiments, the dielectric film stack

142 of is a layered structure of silicon oxynitride films.

(61) In an embodiment, the dielectric stack **142** has a repeating stack **142a** of two dielectric films in which the constituent films within the repeating stack structure **142a** are of differing refractive indices. Differences in the refractive indices can occur primarily from changes in the stoichiometric composition of the films. In preferred embodiments, the changes in the stoichiometry of the films in the repeating film structure **142** is accomplished with changes in the process conditions used in the deposition of the films in the repeating film structure **142**. In a preferred embodiment, the repeating stack structure **142a** includes a first film **143** of 900 nm of silicon oxynitride with an index of refraction of 1.6 and a second film **144** of 50 nm of silicon oxynitride with an index of refraction of 1.7. In another preferred embodiment, the repeating structure **142a** includes a first film **143** of 40 nm of silicon oxynitride with an index of refraction of 1.7 and a second film **144** of 500 nm of silicon oxynitride with an index of refraction of 1.65. In yet another preferred embodiment, the repeating structure **142a** includes a first film **143** of 60 nm of silicon oxynitride with an index of refraction of 1.7 and a second film **144** of 500 nm of silicon oxynitride with an index of refraction of 1.65. It is to be understood that the order of the first film **143** and the second film **144** in embodiments can be reversed and remain within the scope and spirit of the invention (FIG. 1B).

(62) In another embodiment, the dielectric stack **142b** has a repeating stack **142** of more than two dielectric films in which the constituent films **145-147** within the repeating structure **142a** are of differing refractive indices, and in some embodiments, of the same or differing thicknesses. In an embodiment, repeating stack **142b** includes a first film **145** of 400 nm of silicon oxynitride with an index of refraction of 1.6, a second film **146** of 500 nm of silicon oxynitride with an index of refraction of 1.65, and a third film **147** of 50 nm of silicon oxynitride with an index of refraction of 1.7 (FIG. 1B).

(63) In yet other embodiments, the repeating stack **142c** of dielectric stack **142** includes more than three layers for which the index of refraction for the constituent layers of silicon oxynitride is varied to achieve the total film thickness of the overall dielectric stack structure **142**. In embodiments, for example, in which the repeating film structure **142a** has two constituent films with a combined thickness of 600 nm, the stack must be repeated 15 times to reach an overall thickness of 9 microns for the dielectric film stack **142**. In other embodiments in which the overall thickness of the dielectric film stack is 9 microns, a repeating stack of 45 constituent layers of 100 nm each can be implemented in which the overall repeating structure **142a-142c** need only be repeated twice to achieve the overall thickness. In yet other embodiments, the repeating structure **142a-142c** of dielectric stack **142** has a layered film structure that does not repeat because the total number of constituent films in the repeating stack provides sufficient overall film thickness for the film structure **142** (FIG. 1B).

(64) In preferred embodiments, the repeating film structure **142** is a composite structure of repeating stacks. In embodiments with the repeating stack **142a**, the overall thickness of repeating film structure **142** is the combined thickness of the repeating stack **142a**, **142b** multiplied by the number of times that the repeating stack **142a-142b** is repeated. For example, the repeating film structure **142a** for a preferred embodiment in which the first layer **143** is 900 nm and the second layer **144** is 50 nm has a total repeating stack thickness of 950 nm and when repeated 9 times, the resulting combined film thickness for dielectric stack **142** is 8590 nm $((900\text{ nm}+50\text{ nm})\times 9=8590\text{ nm})$. Similarly, in another preferred embodiment, the repeating film structure **142a**, which has a first layer **143** that is 40 nm with a refractive index of 1.7, and which has a second layer **144** that is 500 nm in thickness with a refractive index of 1.65, has a combined thickness for repeating stack **142** of 540 nm, and when repeated 10 times, has a resulting combined film thickness for dielectric stack **142** of 5400 nm $((500\text{ nm}+40\text{ nm})\times 10=5400\text{ nm})$ (FIG. 1B).

(65) Generally, the overall dielectric stack **142** is made sufficiently thick to provide the low optical loss for optical signals transmitted through the resulting waveguide structure **140**. The multilayer

structure, deposited at low temperatures, ensures low stress in the resulting film structure and enables thick waveguides (2000 nm to 25000 nm) to be formed. Waveguide structures **140** are thus sufficiently thick to enable transmission of the optical signals with little interaction of the transmitted optical signals with the substrate, interaction levels that could lead to undesired attenuation of the transmitted signals.

(66) It is to be understood that the thickness, the number of films, and the refractive index for the films in dielectric stack **140** can vary and remain within the scope of the current invention. The refractive index of silicon oxynitride films can vary in the range of 1.4 to 2.02. As the concentration of nitrogen in deposited silicon oxynitride films is minimized, the refractive index approaches the index of refraction of silicon dioxide, 1.445. Conversely, as the concentration of oxygen is minimized in the deposited films, the refractive index approaches the index of refraction of silicon nitride, 2.02. The index of refraction can thusly be varied in the range of 1.445 to 2.02 by varying the stoichiometric concentration of silicon, oxygen, and nitrogen in the deposited films. In embodiments, the index of refraction for the constituent films **143**, **144** in the repeating dielectric film stack **142a**, for example, are varied in the range of 1.445 to 2.02 to produce thick film structures of 2000 to 25000 nm, or greater, and that provide low stress and low optical signal losses, in dielectric film stacks **140**.

(67) In another preferred embodiment, the dielectric film stack **142** includes a repeating stack **142a** with a first layer **143** of silicon oxynitride with thickness of 60 nm and an index of refraction of 1.7 and a second layer **144** of silicon oxynitride with thickness of 500 nm and an index of refraction of 1.65. Repeating dielectric stack structure **142a** is repeated in an embodiment 13 times for a total thickness for dielectric film stack **142** of 7280 nm. It is to be understood that the total number of repeating film stacks **142a** can vary. In some preferred embodiments, the number of repeating film stacks **142a** is three to twenty. In some other preferred embodiments, the repeating film stack **142a** is such to produce a total dielectric film structure **142** that in some embodiments is greater than 2000 nm in thickness and in some embodiments less than 25000 nm. In yet other preferred embodiments, the total dielectric film structure **142** is in the range of 8000 to 12000 nm. In yet other embodiments, the number of repeating film stacks **141** is two or more and the thickness of the dielectric film structure **142** is greater than 2000 nm and less than 25000 nm (FIG. 1B).

(68) In some embodiments, the thickness for the first film **143** is in the range of 5 nm to 1000 nm. In some other embodiments, the thickness of the second film **144** is in the range of 5 nm to 1000 nm. In these and other embodiments, the thickness of the dielectric film structure **142**, which is the sum of the thicknesses of the repeating film structures **142a**, is greater than 2000 nm in thickness. In preferred embodiments, the thickness of the sum of the repeating film structures **142a** is in the range of 4000 to 10000 nm.

(69) It is to be understood that the repeating film structure **142a** is an integral component of the inventive dielectric stack structure **140**. It is also to be understood that the number of films, the film thicknesses, the refractive indices, and the resulting composition of the films can be varied and remain within the spirit and scope of the inventive dielectric stack structure **140**, and in the practice of utilizing the dielectric stack structure **140** to provide low stress and low optical loss for signals transmitted through waveguides that are fabricated from the dielectric stack structure **140**. In this regard, in some embodiments, an initial repeating film structure **142a** is used for two or more of the films in the dielectric stack **142**, and then a different repeating film structure **142a** is used for another two or more films in the same dielectric film structure **140** to produce inventive dielectric stack **140**. It is to be further understood that an initial repeating film structure **142a** can be used for two or more of the films in the dielectric film structure **142**, a different repeating film structure **142a**, can be used for another two or more films in the same dielectric film structure **142**, and then any number of additional repeating film structures **142a** with the same or different repeating film structures can be used for two or more additional films in the dielectric film structure **140** and remain within the scope and spirit of the embodiments. In the foregoing discussion, the variations

in the first film **143** and second film **144** can be produced with one or more variations in the refractive index, the thickness, and the composition or stoichiometry of the films (FIG. 1B).

(70) It is also to be understood that in some embodiments, first film **143** in the repeating film structure **142a** can include one or more films and remain within the scope of the invention. In an embodiment, first film **143** in repeating film structure **142a**, for example, is 500 nm in thickness with a refractive index of 1.7. In another embodiment, first film **143** includes a first part that is 250 nm in thickness with a refractive index of 1.7 and a second part that is 250 nm in thickness with a refractive index of 1.65. In yet another embodiment, the first film **143** in the repeating film structure **142a** has a refractive index of 1.68 with a first partial thickness that is 250 nm and a second partial thickness that is deposited in a separate process step from the first, for example, and that is also 250 nm in thickness for a combined thickness of 500 nm for the two partial films of the first film **143** of repeating film structure **142a** (FIG. 1B).

(71) In some embodiments, the first film **143** has a graded refractive index or stoichiometric composition. Gradations in the composition of the first film **143** of the repeating film structure **142a**, for example, remain within the scope of the current invention. In some embodiments, the refractive index varies through part or all of the thickness of the first film **143**. Similarly, in some embodiments, the stoichiometric composition varies through part or all of the thickness of the first film **143**. Variations in the refractive index or the stoichiometric composition of the first film **143** within the thickness of this film remain within the scope of the current invention (FIG. 1B).

(72) It is also to be understood that in some embodiments, second film **144** in the repeating film structure **142a** can include one or more films and remain within the scope of the invention. In an embodiment, second film **144** in repeating film structure **142a**, for example, is 500 nm in thickness with a refractive index of 1.7. In another embodiment, second film **144** includes a first part that is 250 nm in thickness with a refractive index of 1.7 and a second part that is 250 nm in thickness with a refractive index of 1.65. In yet another embodiment, the second film **144** in the repeating film structure **142a** has a refractive index of 1.68 with a first partial thickness that is 250 nm and a second partial thickness that is deposited in a separate process step from the first, for example, that is also 250 nm for a combined thickness of 500 nm for the two partial films of the second film **144** of the repeating film structure **142a** (FIG. 1B).

(73) In some embodiments, the second film **144** has a graded refractive index or stoichiometric composition. Gradations in the composition of the second film **144** of the repeating film structure **142a**, for example, remain within the scope of the current invention. In some embodiments, the refractive index varies through part or all of the thickness of the second film **144**. Similarly, the stoichiometric composition varies through part or all of the thickness of the second film **144**. Variations in the refractive index or the stoichiometric composition of the second film **144** within the thickness of this film remain within the scope of the current invention (FIG. 1B).

(74) In some embodiments, repeating structure **142** has an unequal number of first layers **143** and second layers **144**. In some embodiments, repeating structure **142** includes a first layer **143** positioned between two second layers **144** (FIG. 1B).

(75) In preferred embodiments, top spacer layer **150** is one or more layers of silicon dioxide or silicon oxynitride. In some embodiments, single spacer layer **150a** is a layer of one type of material, such as silicon dioxide. In some preferred embodiments, single spacer layer **150a** is a layer of silicon oxynitride. In yet other preferred embodiments, the single spacer layer **150a** is a layer of silicon oxynitride with refractive index of 1.55 and with a thickness of 500 nm. In yet other embodiments, single spacer layer **150a** is a layer of silicon oxynitride with thickness in the range of 0 to 1000 nm. Although in preferred embodiments, a spacer layer **150a** is included in the structure, in some other embodiments, the spacer layer **150** can be combined with an optional top layer, can be made very thin, or is not included (FIG. 1A).

(76) Spacer layer **150** can be a composite spacer layer **150b** of one or more layers of silicon oxynitride or silicon dioxide. In an embodiment, composite spacer layer **150b** includes two layers

of silicon oxynitride with thicknesses of 250 nm and with a composite refractive index of approximately 1.55. In some embodiments, the sum of the thicknesses of the two layers in composite spacer layer **150b** is in the range of 1 to 1000 nm (FIG. 1A).

(77) Similarly, the spacer layer **150** can be a composite layer **150c** of three or more layers with the same or different thicknesses and refractive indices, that when combined, provide a total thickness in the range of 1 nm to 1000 nm and a composite refractive index in the range of 1.4 to 2.02 (FIG. 1A).

(78) Optional top layer **158** is one or more layers of a dielectric material such as silicon dioxide, silicon nitride, aluminum oxide, and aluminum nitride, among others. In some embodiments, a top layer **158** of silicon dioxide with thickness of 200 nm and a refractive index of 1.445 is used. In some embodiments, the film thickness of the top layer is in the range of 0 to 500 nm. In some embodiments, silicon oxynitride is used in the optional top layer **158**. In some embodiments, another dielectric material or combination of materials such as aluminum nitride or aluminum oxide is used. In some embodiments, no optional top layer **158** is provided.

(79) The advantages of the current invention with regard to achievable ranges of the measured film stress for films that can be implemented in fabricating dielectric film structures are shown in FIG. 2 for some embodiments. In FIG. 2A, the measured film stress is shown for a range of thicknesses for the inventive dielectric film stacks. FIG. 2A shows that the film stress can be controlled to less than approximately 20 MPa for embodiments as thick as approximately 18 μm . These relatively low stress levels are not achievable or very difficult to achieve in films of a single thick layer of material such as silicon dioxide or silicon oxynitride. In FIG. 2B, the measured stress levels for deposited silicon oxynitride films are shown for films of various refractive indices. As shown, the refractive index is a convenient means for assessing variations in film properties for deposited films. The capability to achieve control of the stress in the individual films over a wide range, allows for the fabrication of very thick dielectric film structures (1000-25000 nm, and greater) with optical properties that are suitable for use as planar waveguides. In embodiments, stress levels are controlled in planar waveguide structures to minimize deformation of the substrates upon which the thick dielectric stacks are deposited, and to achieve low optical signal loss in waveguides fabricated from these thick dielectric film structures.

(80) Referring to FIG. 3, the measured optical losses from some embodiments of the inventive dielectric stack structures are shown. Optical signal losses for practical use in planar waveguide structures of less than approximately 1 dB/cm are desirable. FIG. 3A shows that these levels are achievable for a range of measured composite refractive indices from the inventive dielectric stack structures. In addition to the properties of the dielectric stack structure itself, the buffer layer also has an influence on the measured losses for optical signals transmitted through waveguides fabricated from the inventive dielectric stack structures. FIG. 3B shows how the thickness of the buffer layer in some embodiments affects the measured optical losses. As the thickness of the buffer layer is increased in these embodiments, the resulting optical losses are reduced to values of much less than 1 dB/cm.

(81) Referring to FIG. 4, steps in the formation of embodiments of the dielectric films and film structures are provided. In FIG. 4A, forming step **400** in the process of forming embodiments of the inventive dielectric stack **140** of silicon oxynitride films at low temperature having low stress and low optical loss is shown. Low temperature in FIG. 4A refers to the temperature of the deposition of the films used in the fabrication of the dielectric stacks, namely less than 500° C. in some embodiments, and in other embodiments, less than or equal to 300° C. Low stress in FIG. 4A refers to stress levels in the deposited films in film structure **140** of less than or equal to approximately 20 MPa, either compressive or tensile. Low optical loss in FIG. 4A refers to optical losses in embodiments of deposited dielectric film stacks **140** of less than approximately 1 dB/cm. The forming step **400** provides for the formation of thick structures of dielectric silicon oxynitride films with low stress, and suitable for use in the transmission of optical signals with low loss.

(82) Referring to FIG. 4B, the forming steps **420** in embodiments for which each individual layer in the inventive dielectric stack **140** of silicon oxynitride films is deposited at low temperature, and with low stress and low optical loss is shown. Low temperature in FIG. 4B refers to the temperature of the deposition of the films used in the fabrication of the dielectric stacks, namely less than 500° C. in some embodiments, and in other embodiments, less than or equal to 300° C. Low stress in FIG. 4B refers to stress levels in the deposited films of less than or equal to approximately 20 MPa, either compressive or tensile. Stress levels of less than 20 MPa in deposited films ensure minimal substrate deformation and reduce the likelihood that the films will delaminate. Low optical loss in FIG. 4B refers to optical losses in embodiments of deposited dielectric film stacks **140** of less than approximately 1 dB/cm. Forming step **420** provides for the formation of thin composite films of dielectric silicon oxynitride deposited sequentially at low temperatures of less than 500° C. to form the thick dielectric stack structures **140** with low stress, and suitable for use in the transmission of optical signals with low loss.

(83) Referring to FIG. 4C, steps in the formation of planar waveguides from a forming step **440** and a patterning step **450** are shown for some embodiments. Formation of the individual dielectric films and the dielectric film structures **440** for the inventive stack structure **140** are shown that include the formation of a dielectric stack of silicon oxynitride films on a substrate **110** with a stack structure that includes a buffer layer **130**, one or more optional bottom spacer layers **138**, a repeating stack of one or more dielectric layers **142**, one or more optional top spacer layers **150**, and an optional top layer **158**. Embodiments for the forming of the dielectric film and film structures **440** utilize one or more of forming step **400** and forming step **420**. Patterning step **450** is combined in embodiments with forming step **440** on the resulting dielectric stack to form one or more planar waveguides from the dielectric stack structures **140**. Patterning steps can include the use of established photoresist patterning processes, in which photosensitive layers are used either directly as a means for transferring a pattern with subsequent dry or wet etch processing, or via a hard mask in which the photoresist is first used to transfer a pattern to a hard mask layer that is then used to transfer the waveguide pattern from the hard mask layer to the dielectric stack layer. Processes for photoresist patterning and subsequent wet and dry etching of film structures are well established for those skilled in the art of dielectric film patterning techniques.

(84) Typical deposition processes for dielectric films used in semiconductor device manufacturing include chemical vapor deposition (CVD), plasma enhanced chemical vapor deposition (PECVD), metalorganic chemical vapor deposition (MOCVD), physical vapor deposition (PVD), atomic layer deposition (ALD), among others.

(85) In embodiments, deposition of the inventive dielectric film structure **140** is accomplished using PECVD. In other embodiments, deposition of the inventive dielectric film structure **140** is accomplished using one or more of the thin film deposition techniques that include chemical vapor deposition (CVD), plasma enhanced chemical vapor deposition (PECVD), metalorganic chemical vapor deposition (MOCVD), physical vapor deposition (PVD), and atomic layer deposition (ALD).

(86) Process parameters such as the process pressure, substrate temperature, and process power levels, as well as the selection of precursor gases and the flows of these and other gases used in the process, among others, can each affect the resulting properties of the deposited films. In the fabrication of nitrogen-containing films such as silicon nitride and silicon oxynitride, for example, precursors such as ammonia gas (NH₃), a source of nitrogen, and silane (SiH₄), a source of silicon, are combined in a plasma environment to facilitate the formation of the thin silicon nitride films. The ammonia and silane can be further combined with one or more oxidizing gases such as oxygen (O₂), nitric oxide (NO), and nitrous oxide (N₂O) to incorporate oxygen into the growing films to form silicon oxynitride. The properties of the silicon nitride and the silicon oxynitride films can be affected by the specific selection of gases in the process mixture and the ratio of the specific gas flows in the mixture. Residual impurities, such as hydrogen, can also affect the properties of the deposited films, particularly through the formation of nitrogen-hydrogen

bonds in the deposited films. In embodiments, a process is provided to fabricate dielectric film structure **140** in which one or more nitrogen precursors that do not contain hydrogen are used in the formation of thin films of silicon nitride and silicon oxynitride to reduce the residual hydrogen concentration in the deposited films.

(87) In embodiments, plasma enhanced chemical vapor deposition (PECVD), as shown for example in the schematic drawing in FIG. 5, is used to deposit silicon oxynitride film structures **140**. PECVD processes are most typically performed under vacuum conditions with precursor gases specific to the types of films that are targeted for deposition. Substrates are most typically heated to elevated temperatures. In the configuration shown in FIG. 5, radio frequency (rf) power is capacitively coupled to the electrode opposite to that on which the substrate resides during the process. In the PECVD system depicted in FIG. 5, substrates reside on the lower electrode during processing, and the rf power is applied to the upper electrode. In embodiments in which the power is applied to the upper electrode, the substrate is not subjected directly to high energy ion bombardment as would be anticipated if the substrate were to be placed on the rf powered electrode.

(88) In an embodiment of the present invention, multilayer dielectric film structure **140** is formed using PECVD technology with a process chemistry that utilizes hydrogen-free nitrogen and oxygen precursors to yield planar waveguides with low optical loss and low stress. PECVD technology is widely used in semiconductor and optical device fabrication. Much of the equipment used in the manufacturing of semiconductor devices can be used in the fabrication of optical devices and the use of the term “semiconductor device fabrication” herein is intended to include optical device fabrication. The fabrication of the optical devices and optical subassemblies, utilize, or have the potential to utilize, the techniques and practices used in semiconductor device fabrication.

(89) In embodiments, the dielectric film stack **140** is deposited onto a substrate using PECVD technology at temperatures in the range of 200-500° C. In other embodiments, the substrate temperature during deposition of the dielectric film structure **140** is in the range of approximately 250-400° C. In yet other embodiments, the dielectric film stack **140** is deposited onto a substrate using PECVD technology at temperatures of approximately 300° C. And in yet another embodiment, the dielectric film stack **140** is deposited at or approximately 300° C. using a gas mixture of silane, nitrogen, and nitrous oxide. In yet another embodiment, the dielectric film stack is deposited using a gas mixture of silane, nitrogen, and oxygen. In yet another embodiment, the dielectric film stack **140** is deposited at or approximately 300° C. using a gas mixture of silane, nitrogen, and nitric oxide.

(90) In general, a gas mixture used in the deposition of silicon oxynitride films must contain at least the primary stoichiometric constituents or elements of the film, namely, silicon, oxygen, and nitrogen. Control of the deposited dielectric film properties is achieved, in part, with the control of the gas flows and mixture ratios of the constituent precursor gases in the deposition system. In an embodiment, for example, silane gas (SiH₄) is used as a precursor to provide the silicon, nitrogen gas (N₂) is used as a precursor for nitrogen, and oxygen gas (O₂) is used as a precursor for oxygen. In embodiments, the independent control of one or more of the gas flows, the ratios of the gases, and the partial pressures of these three gases in the deposition system can provide a means for independent control of the ratio of the three elements, namely Si, N, and O, in the deposited films.

(91) In embodiments, silane gas, one of the most commonly used precursors for the deposition of silicon and silicon-containing thin films in epitaxial and chemical vapor deposition processes in semiconductor device fabrication, is provided as a source for silicon in the deposited film structure **140**. In other embodiments, one or more of a silicon precursor that includes dichlorosilane, trichlorosilane, methylsilane, silicon tetrachloride, chlorosilane, dichlorosilane, and trichlorosilane is utilized as a silicon precursor in the deposited dielectric film structure **140**.

(92) In embodiments, a silicon-containing precursor gas that may or may not contain hydrogen, is

combined with a hydrogen-free precursor gas for nitrogen, and a hydrogen-free precursor gas for oxygen to produce the oxynitride layers in the dielectric stack structure **140**. In other embodiments, silane (SiH_4) is used as the silicon precursor in the PECVD deposition of silicon oxynitride films. In yet other embodiments, silane is used as the silicon precursor and is mixed with hydrogen-free precursors for nitrogen and oxygen to form films of silicon oxynitride.

(93) In addition to the source of silicon in one or more of the precursors in the deposition gas mixture, precursors for embodiments with silicon oxynitride in the dielectric film structure **140** include one or more of an oxygen-containing precursor, a nitrogen-containing precursor, and a precursor that contains both oxygen and nitrogen. Precursors that contain both oxygen and nitrogen elements in embodiments include nitrous oxide, nitric oxide, nitrogen dioxide, dinitrogen tetraoxide, and mixtures of these gases, for example. In other embodiments, a nitrogen-containing precursor that does not contain oxygen is used in combination with an oxygen-containing precursor that does not contain nitrogen. In an embodiment, nitrogen gas (N_2) is the nitrogen-containing precursor and is combined with oxygen gas (O_2) as the oxygen-containing precursor, and further combined with silane or another silicon-containing precursor described herein in a plasma enhanced chemical vapor deposition system. In yet other embodiments, a nitrogen-containing precursor such as nitrogen gas is combined with one or more precursors that contain both oxygen and nitrogen, such as nitrous oxide and nitric oxide, or a mixture thereof, and further combined with silane or another silicon-containing precursor as described herein in a plasma enhanced chemical vapor deposition system to form the silicon oxynitride layers in dielectric stack **140**. In yet other embodiments, an oxygen-containing precursor such as oxygen gas is combined with one or more precursors that contain both oxygen and nitrogen, such as nitrous oxide and nitric oxide, and further combined with silane or another silicon-containing precursor in a plasma enhanced chemical vapor deposition system or other deposition system as described herein to form dielectric stack **140**. Oxygen-containing precursors include atomic oxygen, molecular oxygen, ozone, carbon monoxide, and carbon dioxide.

(94) And in yet other embodiments, one or more of argon, helium, neon, xenon, nitrogen, and oxygen is added to the gas mixtures described herein in embodiments as a diluent. Diluents are often utilized in semiconductor deposition processes to slow the deposition rate, to enhance the ignition of the plasma in plasma-based processes, to improve deposition uniformity, and to alter or modify the energy absorption profile of the gaseous species in the plasma, among other potential benefits. Addition of one or more diluents listed herein to the processes described herein remains within the scope of the current invention.

(95) In embodiments, the use of hydrogen-free, nitrogen and oxygen precursors in the deposition of the inventive dielectric film stacks yields low stress silicon oxynitride film structures from which waveguides can be formed that exhibit low optical signal loss.

(96) In other embodiments, low stress, low optical loss silicon oxynitride film structures are formed using plasma enhanced chemical vapor deposition with a gas mixture that contains one or more of each of a silicon-containing precursor, a nitrogen-containing precursor, and an oxygen-containing precursor.

(97) Referring to FIG. 6, a plot of the refractive index is shown for silicon oxynitride films deposited using such a gas mixture in a PECVD system. The ordinate in the plot in FIG. 6 shows the flow rate of a molecular precursor gas that contains silicon and the abscissa shows the refractive indices for a measurement wavelength of 633 nm that correspond to specific values of the precursor gas flow rate used in the deposition of the inventive dielectric stack structures **140**. The plot includes sample data that show the variation in a representative optical property, namely the refractive index, for silicon oxynitride films that were deposited over a range of gas flows for the silicon-containing precursor, holding other gas flows and PECVD system parameters constant.

(98) The data provided in FIG. 6, when combined with the measured stresses over a similar range of refractive indices as provided in FIG. 2B, demonstrate the benefits of the inventive process as a

means to provide control of the optical properties (refractive index) and mechanical properties (stress) of the deposited films with variation of the gas flow mixture. The range of gas mixtures shown in the exemplary sample data shown in FIG. 6, produced with variations in the silicon-containing precursor gas flow, provide a wide range of measured refractive indices for silicon oxynitride films. The variation in the measured refractive index are shown in FIG. 2B to produce the low values of residual stress in deposited silicon oxynitride films, and that a crossover point of zero stress can be produced a specific value of the silicon-containing precursor gas flow. The innovation provided using the inventive processes should become increasingly apparent in embodiments in which the achievable optical and mechanical properties are further combined with the inventive superlattice stack structures described herein for the dielectric stack structures **140** as described in FIG. 1. Combinations of refractive index and film thickness yield stack structures in embodiments that have low stress and low optical loss when deposited with processes that do not contain hydrogen in the nitrogen and oxygen precursors.

(99) In embodiments of the inventive PECVD processes, a dielectric stack structure **140** is provided on substrate **100** wherein the ratios of the precursor gases in the PECVD process are configured to achieve a stress level of less than approximately 20 MPa in magnitude. In other embodiments, the dielectric stack structure **140** is a structure of silicon oxynitride layers deposited in a PECVD system using a process that contains silane, nitrogen, and nitrous oxide wherein the ratios of these precursor gases are configured to achieve a stress level that is less than or approximately 20 MPa. In embodiments, control of the stress is achieved with the ratios of the gases in combination with the film thicknesses of the individual layers in the dielectric film structure **140**, as described in examples provided herein.

(100) In embodiments of the dielectric stack structures **140**, for example, the stoichiometric concentrations of the individual layers are selected, as reflected in the measured refractive index of the individual layers in the stack structure, and optimized along with the layer thicknesses to produce low stress film structures with low optical signal loss characteristics. In the formation of the stacked dielectric structure **140**, the total composite refractive index and the stress level in the film stack **140** are affected by the number of layers, the characteristics of the interfaces between the layers, the stoichiometric concentration, the resulting density of the deposited films, and the presence of impurities in the deposited films. This suggests that the measured refractive index may not be unique. Although the refractive index provides an effective method for characterizing the effect of changes in process parameters such as the gas flow on the optical properties of the deposited dielectric films, multiple stoichiometric concentrations may exist for the elemental constituents of the silicon oxynitride layer for a given value of refractive index. Nonetheless, the refractive index is an effective means for providing a measure of changes in the film properties, and most notably, a measure of the residual stress in the deposited films.

(101) The data shown in FIG. 6 is provided for demonstrative purposes in that the process parameters such as plasma power, gas pressure, substrate temperature, and gas ratios in the PECVD system can influence the measured refractive index. The effects of the variation in the refractive index are likely a reflection of the variation in the stoichiometric concentrations and perhaps the density of the deposited films. It is important to note, therefore, that the refractive index may not be unique to a single stoichiometric combination of silicon, oxygen, and nitrogen in the silicon oxynitride layers.

(102) Variations in the embodiments in which the inventive dielectric stack structure **140** is combined with the silicon, oxygen, and nitrogen precursors as described herein remain within the scope of the current invention.

(103) In embodiments, the low stress, low optical loss dielectric film structures **140** are formed using plasma enhanced chemical vapor deposition with a gas mixture that does not include hydrogen-containing precursors for either nitrogen, oxygen, or both, or that includes concentrations of these gases that are low enough so as to not require high temperature processing of greater than

500° C. for example, or high thermal budget processes, to achieve the low optical signal losses of less than 1 dB/cm in waveguides fabricated from the inventive dielectric stack structure **140**. In some embodiments, a small amount of hydrogen-containing nitrogen or oxygen precursor gas, or both, of approximately 5-10% or less of the total precursor gas flow, is added to the PECVD gas mixture. In these embodiments, the concentrations of hydrogen-containing nitrogen and oxygen precursors are low enough so as to not require high thermal budget processes to achieve the low optical signal losses of less than 1 dB/cm in waveguides fabricated from the inventive dielectric stack structure **140**. In yet other embodiments, a nitrogen-containing precursor such as nitrogen gas is combined with a small amount of a hydrogen-containing precursor gas, such as ammonia, for example, and one or more precursors that contain both oxygen and nitrogen, such as nitrous oxide and nitric oxide, and further combined with silane or another silicon-containing precursor as described herein in a plasma enhanced chemical vapor deposition system to form the silicon oxynitride layers in the dielectric stack **140**. In these embodiments, the amount of ammonia in the process should be sufficiently low, less than approximately 10% of the total gas flow for example, so as to not require high temperature processing of greater than 500° C. for example, or processing with high thermal budget processes, to achieve the low optical signal losses of less than 1 dB/cm in waveguides fabricated from the inventive dielectric stack structure **140**. Although increases in the optical signal loss are to be expected, the inclusion of small amounts of ammonia to the gas mixture will increase the deposition rate of the silicon oxynitride films.

(104) FIG. 2B shows the variation in the measured stress levels in deposited silicon oxynitride films over a range of measured refractive index for these films. The data in FIG. 2B show that the stress in the deposited films using the inventive process can be varied over a wide range between compressive stress and tensile stress with the crossover at 0 MPa in the measured stress observed at the transition between the films being in tension and compression. In embodiments, the stress in each layer in the inventive stack structures is varied such that the total stress in the dielectric film structure as shown in FIG. 1 is less than approximately 20 MPa. The stresses in each of the layers of the inventive stack structure as described in FIG. 1, is controlled in embodiments by varying the gas ratio, as for example as shown in FIG. 2B, and by varying the corresponding thickness of the layers. Examples of combinations of specific refractive indices, produced with variations in the nitrogen to nitrous oxide gas ratio, are described herein.

(105) In embodiments, when the use of hydrogen-free nitrogen and oxygen precursors is combined in the deposition chemistry with substrate temperatures during deposition of less than 400° C. in embodiments, and of approximately 300° C. in yet other embodiments, the resulting film stacks can be produced that exhibit stress levels of less than 20 MPa. Additionally, in embodiments, the inventive film stacks are patterned to form optical waveguides that provide low losses for optical signals traveling within these waveguides. Examples of embodiments of the deposited dielectric film stacks are described herein, and in particular are described in the relevant discussion of FIG. 1 described herein.

(106) In embodiments, the stack of silicon oxynitride films is formed onto a substrate using a PECVD process for which the process parameters, including the precursor gas ratios, are configured to achieve a stress level of less than 25 MPa, and in preferred embodiments less than 20 MPa. Measured stress levels in the deposited film stacks of less than 20 MPa greatly reduce the potential for either the substrate to deform, the deposited films to delaminate, or for some other undesired effect to manifest in either the waveguide or the finished subassembly that includes the waveguide. Substrate deformation and film delamination are just two forms of damage that can occur when stresses in the deposited film structures are not properly controlled. In embodiments, the deposited dielectric film stacks with stress levels of less than 20 MPa are patterned into one or more of waveguides and optical devices.

(107) In yet other embodiments, a stack of silicon oxynitride films are formed onto a substrate using a PECVD process for which the gas ratios are configured to achieve a stress level of less than

20 MPa and for which the process chemistry includes a silicon-containing precursor that may or may not contain hydrogen, and one or more molecular precursors that contain nitrogen or oxygen elements that contains little or no ammonia or other hydrogen-containing gas. In embodiments, the deposited film stacks with stress levels of less than 20 MPa that are deposited using non-hydrogen-containing nitrogen and oxygen gas sources are patterned into one or more of waveguides and optical devices.

(108) In yet other embodiments, a stack of silicon oxynitride films are formed onto a substrate using a PECVD process for which the gas ratios are configured to achieve a stress level of less than 20 MPa and for which the process chemistry does not include ammonia or other hydrogen-containing gas, other than silane. Silane (SiH_4) is widely used in industry for the deposition of silicon-containing films and is used in some embodiments. In embodiments, the deposited film stacks with stress levels of less than 20 MPa, that are deposited using non-hydrogen-containing nitrogen and oxygen gas sources such as nitrogen, oxygen, nitric oxide, and nitrous oxide, for example, are patterned into one or more of waveguides and optical devices.

(109) In embodiments, the gas pressure in the PECVD system during the deposition of the silicon oxynitride films can be in the range of 1 mTorr to 100 Torr to produce dielectric film stack structures. In other embodiments, process pressures are in the range of 50 mTorr to 2 Torr. In yet other embodiments, process pressures are in the range of 100 to 5000 mTorr. The process pressure need not be the same for every step in the deposition sequence required to produce the full dielectric film structure **140**.

(110) In embodiments, the total gas flows during deposition of inventive dielectric stack structure **140** used in the inventive dielectric interposers can vary in the range of 5 sccm to 5000 sccm. In some embodiments, the silane gas flow is in the range of 5-1000 sccm. In other embodiments, the silane gas flow, or other silicon-containing precursor gas, is in the range of approximately 10-100 sccm.

(111) In embodiments, the nitrogen and oxygen precursor gas flows can vary over a wide range. In some embodiments, the nitrogen and oxygen precursor gas flows are in the range of 0 to 5000 sccm.

(112) In some embodiments, for which nitrogen is combined with nitrous oxide or nitric oxide, the nitrogen gas flow and nitrous oxide gas flows are in the range of 0 to 5000 sccm. And in some other embodiments, for which oxygen is combined with nitrous oxide or nitric oxide, the oxygen gas flow and nitrous oxide gas flows are in the range of 0 to 5000 sccm.

(113) In a PECVD system, the input power to the plasma can vary over a wide range. Typical input power for the dielectric stacks in embodiments is approximately 25-2000 W. In some embodiments, the input power during plasma enhanced deposition steps is in the range of 200 to 700 W. In yet other embodiments, the input power during plasma enhanced deposition steps is in the range of 500 to 1200 W. The process power need not be kept constant for the deposition of each of the films in the dielectric stack structure **140** but can vary from step to step, or within a deposition step.

(114) In embodiments, one or more frequencies in the range of 1 kHz to 1 GHz can be used for the rf power provided to the plasma. In an embodiment, a frequency of 13.56 MHz is used to generate the plasma. In other embodiments, other frequencies or combinations of frequencies are used to generate the plasma. In some other embodiments, rf power with a frequency of approximately 27 MHz is applied to the powered electrode. In a preferred embodiment, 13.56 MHz rf power is applied to the upper electrode with the substrate residing on a grounded electrode during the deposition of the dielectric film structure **140**. In other embodiments, the electrode upon which the substrate resides during the deposition process is not grounded. In yet other embodiments, the power is applied to both the upper and lower electrodes, divided either equally or unequally.

(115) Power can be delivered to the PECVD system in either capacitive or inductive operational modes and remain within the scope of the current invention. For inductive-coupled PECVD systems, the plasma is typically generated with power that is delivered to the plasma via an antenna

and the wafer typically resides on a bottom electrode similar to that shown in the capacitively coupled configuration shown in FIG. 5.

(116) Inert gases, such as argon and helium, can be added to the gas mixture, to yield dielectric film stacks with similar properties and remain within the scope of the current invention. Inert gases are frequently added to deposition process chemistries to modify film properties, such as the stoichiometry or density of the film. In embodiments, argon is added to a gas mixture of silane, nitrous oxide, and nitrogen. In yet other embodiments, helium is added to the gas mixture of silane, nitrous oxide, and nitrogen. In yet other embodiments, one or more of argon and helium is added to a process gas mixture that includes a silicon-containing precursor gas that may or may not contain hydrogen, a nitrogen precursor that does not include hydrogen in the nitrogen-containing precursor gas molecule or molecules, and an oxygen precursor gas that does not contain hydrogen in the oxygen-containing precursor gas molecule or molecules. Examples of nitrogen precursors that do not include hydrogen include molecular nitrogen (N_2), nitrous oxide (N_2O), and nitric oxide (NO). Examples of oxygen precursors that do not include hydrogen include molecular oxygen (O_2), nitrous oxide (N_2O), nitric oxide (NO), carbon monoxide (CO), and carbon dioxide (CO_2).

(117) Commercially available deposition systems such as the APM PECVD model manufactured by SPTS provide advanced programmability for the deposition of the overall dielectric film structure **140** and for each step in the deposition process. The programmability of commercially available deposition systems enables automated operation of the system hardware that includes, for example, the mass flow controllers, the source power supply or supplies, the temperature controllers, and the pressure controllers. The process parameters such as, for example, the gas flow rates, the power level or levels, the pressure, the substrate temperature, among other parameters for each step in the deposition of a film structure can be programmed into a process recipe. Use of programmable and commercially available deposition systems is widely used in the industry and the use of programmable deposition systems is anticipated and within the scope of the current invention.

(118) Referring to FIG. 7A, a cross sectional schematic of an embodiment of the inventive optical dielectric interposer structure **500** is shown. In this figure, an embodiment for interposer **700** includes substrate **710**, optional interconnect layer **720**, and planar dielectric stack structure **740** disposed on the optional interconnect layer **720**. Terminal pad opening **725** in the interconnect layer **720** provides for connections of optical die to the interconnect metal lines. In some embodiments, the top intermetal dielectric **727** in the interconnect layer resides below the dielectric stack **740** as shown in FIG. 7A. The interconnect layer **720** is a structure of metal lines **726** and intermetal dielectric films **727** that provide metal traces for mounting optical devices and for interconnecting electrical and optoelectrical die on the dielectric interposer **700**. In some embodiments, the top layer of the interconnect layer **720** may be electrically conductive or insulating, or may be electrically conducting in some areas and insulating in some areas. In preferred embodiments in which optical, electrical, or optoelectrical die are mounted onto the interposer **700**, metal traces are routed within the interconnect layer **720** that are accessible through openings **725** to provide electrical and mechanical connections for the optical, electrical, and optoelectrical devices in, on, or connected to the interposer **700**. It is to be understood that the mounting of purely optical die (i.e., die that have an optical function but that are not electrical) as in a discrete waveguide for example, can benefit from the methods of mechanical attachment commonly used in the attachment of electrical die. Attachment of purely optical devices using electrical bond pads is within the scope of the current invention as described herein. It is also important to note that the top layer of the intermetal dielectric **727** can provide the same functionality as the buffer layer **730** in some embodiments as shown in FIG. 7A.

(119) The inclusion of optical, electrical, and/or optoelectrical devices, forms a submount assembly **705** from the inventive optical dielectric interposer **700**. FIG. 7B shows a cross sectional schematic

of an embodiment of a submount assembly **705** with an optical fiber **790** positioned to provide an optical pathway for the transmission of optical signals between the optical fiber **790** and the planar dielectric stack **740**. FIG. 7B also shows optical, electrical, or optoelectrical device **760** and electrical device **762** mounted to terminal pad openings **725** in interconnect layer **720**. In an embodiment, optical signals are received from optical fiber **790** into a waveguide fabricated from the planar dielectric stack **740** and routed to device **760** for processing, re-routing, or conversion to electrical signals, for example.

(120) In other embodiments, the optical signals originate on the submount assembly **705** and are transmitted through waveguides fabricated from planar dielectric stack structure **740** to the optical fiber **790**. In yet other embodiments, the signals are both received from, and transmitted to, the optical fiber **790**.

(121) Referring to FIG. 8, cross sectional schematics of embodiments of the inventive optical dielectric interposer structure **800** and the submount assembly **805** are shown. In FIG. 8A, an embodiment for interposer **800** includes substrate **810**, optional interconnect layer **820**, and planar dielectric stack structure **840** disposed on the optional interconnect layer **820**. Interconnect layer **820** is typically provided in embodiments for which interconnects are required for optical or electrical die mounted on the interposer **800** to form a submount assembly. Terminal pad opening **825** in the interconnect layer **820** provides connections for the optical and electrical die to the interconnect metal lines **826**. Interconnect metal lines **826** within interconnect layer **820** form interconnects between electrical devices mounted onto the interposer **800**, and in some embodiments, to form electrical connections for devices external to the interposer **800**. In embodiments, the planar dielectric stack **840** includes buffer layer **830**. In some other embodiments, openings in the buffer layer **830** provide access to underlying metal layers **826** through the interconnect layer openings **825**. It is to be understood that the buffer layer **830** can be utilized for multiple purposes on the interposer **800** that include isolation, insulation, vertical spacing, alignment, and control of optical loss. In some embodiments, the patterning of the buffer layer **830** is not coincident with the pattern of the other layers in waveguides that are fabricated from the inventive dielectric stack structure **840**. In yet other embodiments, the buffer layer can be a part of the intermetal dielectric **827** of the interconnect layer **820**.

(122) In embodiments, the intermetal dielectric **827** in the interconnect layer **820** generally provides electrical isolation for the metal interconnects **826**. The interconnect layer **820** is a structure of metal lines **826** and intermetal dielectric **827** that provide insulated electrical interconnections for the electrical and optoelectrical die on the dielectric interposer **800**, and in some embodiments, allow for the interconnection of devices mounted external to the interposer **800** but for which connections are required within the interposer **800**. It is understood that optical devices that do not require electrical interconnection can also be attached in some embodiments to interconnect layers for the purpose of mechanical attachment without a specific requirement for electrical interconnection.

(123) The inclusion of electrical, optical, and/or optoelectric devices forms a submount assembly **805** from the optical interposer **800** on substrate **810** with interconnect layer **820**. In FIG. 8B, a cross sectional schematic of an embodiment of a submount assembly **805** with an optical fiber **890** positioned to provide an optical pathway between the optical fiber **890** and planar dielectric stack **840** is shown. FIG. 8B also shows optoelectrical device **860** and electrical device **862** mounted through buffer layer **830** to terminal pad openings **825** and connected to metal interconnect lines **826** in interconnect layer **820**. In embodiments, intermetal dielectric **827** provides electrical insulation for the metal interconnects **826**. In an embodiment, optical signals are received from optical fiber **890**, are directed into planar waveguides fabricated from inventive dielectric stack **840**, and routed to aligned optical or optoelectrical device **860** for processing, re-routing, or conversion to electrical signals, for example.

(124) Referring to FIG. 9, cross sectional schematics of embodiments of the inventive optical

dielectric interposer **900** and submount assembly **905** are shown. In FIG. **9A**, an embodiment for interposer **900** includes substrate **910**, interconnect layer **920**, inventive planar dielectric stack structure **940** disposed on interconnect layer **920**, and integrated electrical device **964**. In some embodiments, integrated electrical device **964** in the underlying substrate **910** is a transistor, capacitor, resistor, inductor, or other electrical device. In other embodiments, integrated electrical device **964** is a p-channel metal oxide semiconductor (PMOS) transistor, an n-channel metal oxide semiconductor (NMOS) transistor device or array of one or more of these devices. In some embodiments, the electrical device **964** is an array of transistor devices based on complementary metal oxide semiconductor (CMOS) technology. In some embodiments, transistor arrays **964** in the substrate **910**, are used for signal processing, signal conditioning, signal generation, memory, and computation, for example. In some embodiments, terminal pad openings **925** in the interconnect layer **920** provide electrical connections between optoelectrical die and the interconnect metal lines **926**. In some embodiments, the top intermetal dielectric **927** in the interconnect layer **920** resides below the dielectric stack **940** as shown in FIG. **9A**, and in some embodiments, the upper layer of the intermetal dielectric **927** can also serve as the buffer layer **930**. The interconnect layer **920** is a structure of metal lines **926** and intermetal dielectric **927** that provide electrical connections for interconnecting electrical and optoelectrical devices and die that are fabricated on, mounted in, or are connected external to the dielectric interposer **900**.

(125) In some embodiments, the top layer of the interconnect layer **920** may be electrically conductive or insulating. Some parts of the top layer of interconnect layer **920** can be insulating, and some parts of the top layer of interconnect layer **920** can be conductive. In preferred embodiments in which electrical or optoelectrical die are mounted onto the interposer **900**, metal lines **926** are routed within the interconnect layer **920** to provide electrical connections for the devices in, on, or connected to the interposer **900**, and to underlying electrical devices **964**.

(126) Submount assembly **905** is formed from the optical dielectric interposer **900** by the inclusion of optical, electrical, and optoelectrical devices **960** onto the interposer **900**. FIG. **9B** shows a cross sectional schematic of an embodiment of a submount assembly **905** with optical fiber **990** positioned to provide an optical pathway between the optical fiber **990** and a waveguide fabricated from the inventive planar dielectric stack **940**. FIG. **9B** also shows optoelectrical device **960** mounted to terminal pad openings **925** on interconnect layer **920**. In embodiments, optical signals are received from optical fiber **990**, into planar waveguides formed from the inventive dielectric stack **940** and routed to optoelectrical or optical device **960** for processing, re-routing, or conversion to electrical signals, for example. In some embodiments, optoelectrical die **960** are connected to one or more of electrical devices **964** via metal lines **926** in the interconnect layer **920**. In these embodiments, the optical signals may also originate, wholly or in part, on the submount assembly **905** from which the signals can be transmitted through the planar waveguide structures **940** to the optical fiber **990**.

(127) In other embodiments, the optical signals originate on the submount assembly **905** and are transmitted through one or more planar waveguide structures formed from the inventive dielectric stack **940** to the optical fiber **990**. In yet other embodiments, the signals are received from the optical fiber **990** for one or more of processing, routing, and conversion to electrical signals.

(128) Referring to FIG. **10**, cross sectional schematics of embodiments of the inventive optical dielectric interposer **1000** and submount assembly **1005** are shown. In FIG. **10A**, an embodiment for interposer **1000** includes substrate **1010**, optional interconnect layer **1020**, inventive planar dielectric stack structure **1040** disposed on the optional interconnect layer **1020**, and integrated electrical device **1064** in substrate **1010**.

(129) In some embodiments, integrated electrical device **1064** in the underlying substrate **1010** is a transistor, capacitor, resistor, inductor, or other electrical device. In other embodiments, integrated electrical device **1064** is a p-channel metal oxide semiconductor (PMOS) or n-channel metal oxide semiconductor (NMOS) device, or array of one or more of these devices. In other embodiments,

electrical device **1064** is an array of transistors based on complementary metal oxide semiconductor (CMOS) technology. In some embodiments, transistor arrays **1064** in the substrate **1010** are used for signal processing, signal conditioning, signal generation, memory, and computation, for example. In some embodiments, the terminal pad opening **1025** in the interconnect layer **1020** provides for electrical connections of optoelectrical die to the interconnect metal lines **1026** in interconnect layer **1020**. In some embodiments, the top layer of the intermetal dielectric **1026** in the interconnect layer **1020** resides below the dielectric stack **1040**. In some embodiments, the planar dielectric stack **1040** includes buffer layer **1030**. In yet other embodiments with buffer layer **1030** in dielectric stack **1040**, the buffer layer **1030** resides within or above the interconnect layer **1020**. Interconnect layer **1020** is typically provided in embodiments for which interconnects are required for optoelectrical die mounted on the interposer **1000** to form a submount assembly **1005**. The interconnect layer **1020** is a structure of metal lines **1026** and intermetal dielectric films **1027** that provide metal connections for interconnecting optical, electrical, and optoelectrical devices and dies that are fabricated on, mounted in, or connected external to the dielectric interposer **1000**.

(130) In some embodiments, the terminal pad openings **1025** in the interconnect layer **1020** provide connections for optoelectrical die **1060** to the interconnect metal lines **1026** as shown in FIG. **10B**. Interconnect metal lines **1026** within interconnect layer **1020** form interconnects between optoelectrical devices **1060** and optional electrical devices (not shown) mounted onto the interposer, or to form connections for one or more of optoelectrical devices and electrical devices connected external to the interposer **1000**.

(131) In some embodiments, the top layer of the interconnect layer **1020** may be electrically conductive or insulating. In preferred embodiments in which optical die are to be mounted onto the interposer **1000**, metal traces **1026** are routed within the interconnect layer **1020** that are accessible through openings **1025** to provide electrical and mechanical connections for the optical, electrical, and optoelectrical devices in, on, or connected to the interposer **1000**, and to the underlying electrical device **1064**. It is to be understood that the mounting of purely optical die (i.e., die that have an optical function but that are not electrical) as in a discrete waveguide for example, can benefit from the methods of mechanical attachment commonly used in the attachment of electrical die. Attachment of purely optical devices using electrical bond pads is within the scope of the current invention as described herein.

(132) In some embodiments, intermetal dielectric **1027** in the interconnect layer **1020** provides electrical isolation for the metal interconnects **1026**. The interconnect layer **1020** is a structure of metal traces **1026** and intermetal dielectric **1027** that provides electrically insulated interconnections for the optical, electrical, and optoelectrical die **1060** on the dielectric interposer **1000**, and in some embodiments, allow for the interconnection of devices mounted external to the interposer **1000** but for which connections are required within the interposer **1000**.

(133) Submount assembly **1005** is formed from the optical interposer **1000** by the inclusion of electrical, optical, optoelectric devices **1060** onto the interposer **1000**. FIG. **10B** shows a cross sectional schematic of an embodiment of a submount assembly **1005** with optical fiber **1090** positioned to provide an optical pathway between the optical fiber **1090** and a waveguide fabricated from the inventive planar dielectric stack **1040**. FIG. **10B** also shows optoelectrical device **1060** mounted to terminal pad openings **1025** in interconnect layer **1020**. In some embodiments, terminal pad openings **1025** are provided through openings in the buffer layer **1030**, or another layer on the surface of the interconnect layer **1020**. In some embodiments, optical signals are received from optical fiber **1090**, into planar waveguides formed from the inventive dielectric stack **1040** and routed to optoelectrical or optical device **1060** for processing, re-routing, or conversion to electrical signals, for example. In some embodiments, optoelectrical die **1060** are connected to one or more electrical devices **1064** via metal lines **1026** in the interconnect layer **1020**.

(134) In other embodiments, the optical signals originate on the submount assembly **1005** and are

transmitted through planar waveguides formed from the inventive dielectric film structure **1040** to the optical fiber **1090**. In yet other embodiments, the signals are received from the optical fiber **1090** to the submount assembly **1005** for one or more of processing, routing, and conversion to electrical signals.

(135) Referring to FIG. **11**, cross sectional schematics of embodiments of the inventive optical dielectric interposer structure **1100** and submount assembly **1105** are shown. In FIG. **11A**, interposer **1100** is shown and includes substrate **1110** and interconnect layer **1120**. Interconnect layer **1120** is a structure of metal traces **1126** and intermetal dielectric material **1127** within which conductive pathways are provided for interconnecting electrical and optoelectrical devices and die that are formed on, mounted in, or connected to the dielectric interposer **1100**. In some embodiments, interconnected devices are interconnected to the interposer **1100** from an external mount or submount assembly. The dotted lines in interconnect layer **1120** shown in FIG. **11A** schematically represent examples of electrical pathways **1126** within the interconnect layer **1120** for interconnecting optoelectrical devices and electrical devices mounted to terminal pad interconnect openings **1125**, for example. FIG. **11A** shows inventive dielectric stack **1140** mounted via bonding pads **1122** as a discrete dielectric waveguide component **1165** to interconnect layer **1120**. In some embodiments, the dielectric stack **1140** is fabricated or formed independently of the substrate **1110** and the interconnect layer **1120**, and then added as a discrete element to form interposer **1100**. It is important to note that the formation of interposer **1100** may be accomplished concurrently with the formation of submount assembly **1105** for embodiments in which the discrete waveguide components **1165**, with inventive dielectric stack **1140**, are added to interposer **1100** concurrently with optoelectrical and electrical components **1160** as shown in FIG. **11B**.

(136) In embodiments, discrete waveguide component **1165**, fabricated with the inventive dielectric stack **1140**, is a simple conduit for the transmission of optical signals. In other embodiments, one or more discrete waveguide components **1165** on submount assembly **1105** are conduits for the transmission of optical signals from an optical fiber attached to the submount assembly to one or more locations on the submount assembly. In yet other embodiments, discrete waveguide components **1165** on submount assembly **1105** are conduits for the transmission and distribution of optical signals from one or more optical fibers attached to the submount assembly to one or more locations on the submount assembly **1105**. In yet other embodiments, discrete waveguide components **1165** on submount assembly **1105** can include one or more of a spot size converter, a filter, an arrayed waveguide, a multiplexers, a demultiplexer, a grating, a power combiner, and the like.

(137) In FIG. **11B**, inventive planar dielectric stack structure **1140** is shown as discrete waveguide component **1145** attached to the interconnect layer **1120** on substrate **1110**. Submount assembly **1105** is formed from the optical dielectric interposer **1100** by the inclusion of optical, electrical, and optoelectrical devices **1160**, **1162** onto the interposer **1100**. FIG. **11B** shows a cross sectional schematic of an embodiment of a submount assembly **1105** with optical fiber **1190** positioned to provide an optical pathway to the discrete dielectric waveguide component **1165**. In the embodiment shown in FIG. **11B**, the inventive planar dielectric stack **1140** is a pre-fabricated discrete optical waveguide component **1165** mounted to interposer **1100**. FIG. **11B** shows optoelectrical device **1160** mounted to terminal pad openings **1125** in interconnect layer **1120** to form submount assembly **1105**. In an embodiment, optical fiber **1190** is aligned to discrete waveguide **1165**, formed from inventive dielectric stack **1140**, which is further aligned to optical device **1160** to allow for the receiving and sending of optical signals for processing, re-routing, or conversion to electrical signals, for example. Optical alignment of devices to the waveguide, in embodiments, provides less than 1 dB power loss, and in preferred embodiments, less than 0.5 dB. Accurate alignment is essential to reducing power loss to tolerable levels.

(138) Terminal pad openings **1125** in the interconnect layer **1120** provide for connections of optoelectrical die **1160** to the interconnect metal traces **1126**. In preferred embodiments in which

optoelectrical die **1160** are mounted onto the interposer **1100**, metal traces **1126** are routed within the interconnect layer **1120** to provide electrical and mechanical connections **1126** for optical, electrical, and optoelectrical devices in, on, or connected to the interposer **1100**. In embodiments, the intermetal dielectric **1127** in the interconnect layer **1120** provides electrical isolation for the metal interconnects **1126**. The interconnect layer **1120** is a structure of metal lines and traces **1126** and intermetal dielectric **1127** that provide interconnections for the optical, electrical, and optoelectrical die **1160**, **1162** on the dielectric interposer **1100**, and in some embodiments, allow for the interconnection of devices mounted external to the interposer **1100** but for which connections are required on or within the interposer **1100**.

(139) In FIG. **11C**, interposer **1100** is shown and includes substrate **1110**, interconnect layer **1120**, discrete waveguide component **1165**, and integrated electrical device **1164**. Interconnect layer **1120** is a structure of metal lines and traces **1126** and intermetal dielectric material **1127** within which conductive pathways for interconnecting electrical and optoelectrical devices **1160**, **1162** that are fabricated on, mounted in, or connected from an external submount assembly to the dielectric interposer **1100**, or provided in underlying substrate **1110**. The dotted lines in interconnect layer **1120** shown in FIG. **11C** schematically represent examples of electrical pathways **1126** within the interconnect layer **1120** for interconnecting optoelectrical devices **1160** and electrical devices **1162** mounted to terminal pad interconnect openings **1125**. In preferred embodiments in which optoelectrical die **1160** are mounted onto the interposer **1100**, metal interconnects **1126** are routed within the interconnect layer **1120** to provide electrical and mechanical connections for electrical and optoelectrical devices in, on, or connected to the interposer **1100**, and to the underlying electrical devices **1164**. Integrated electrical device **1164** in underlying substrate **1110**, in some embodiments, is one or more of a transistor, capacitor, resistor, inductor, or other electrical device, or array of electrical devices. In other embodiments, integrated electrical device **1164** is a p-channel metal oxide semiconductor (PMOS) transistor or an n-channel metal oxide semiconductor (NMOS) device, or array of one or more of these devices. In yet other embodiments, device **1164** is an array of transistors based on complementary metal oxide semiconductor (CMOS) transistor technology. In yet other embodiments, the integrated electrical device **1164** is a bipolar transistor or an array of bipolar transistor devices. In yet other embodiments, the integrated electrical device **1164** is a field effect transistor or an array of field effect transistors. In some embodiments, transistor arrays **1164** in the substrate **1110**, are used for signal processing, signal conditioning, signal generation, memory, and computation, for example.

(140) In FIG. **11C**, the inventive dielectric stack **1140** is shown in the form of a discrete dielectric waveguide component **1165** mounted to interconnect layer **1120** via bonding pads **1122**. In some embodiments, the dielectric stack **1140** is fabricated independently of the substrate **1110** and the interconnect layer **1120**, and then added as a discrete element to form interposer **1100** as shown, for example, in FIG. **11C**. Although electrical connections are not required for optical waveguides, bonding pads **1122**, in some embodiments, are similar to bond pads used to form electrical connections. In other embodiments, other adhesion methods are used that include adhesive, epoxy, or other bonding material.

(141) Submount assembly **1105**, shown in FIG. **11D**, is formed from the optical interposer **1100** with the inclusion of electrical, optical, optoelectric devices on the interposer **1100**. It is important to note that the formation of inventive interposer **1100** with the addition of the discrete waveguide **1165** is accomplished concurrently with the formation of submount assembly **1105** for embodiments in which the discrete waveguide components **1165** are added to interposer **1100** concurrently with optical, optoelectrical, and electrical components **1160**. FIG. **11D** shows a cross sectional schematic of an embodiment of a submount assembly **1105** with optical fiber **1190** positioned to provide an optical pathway between the optical fiber **1190** and a planar waveguide **1165** fabricated from the inventive planar dielectric stack **1140**. FIG. **11D** also shows optoelectrical device **1160** mounted to terminal pad openings **1125** in interconnect layer **1120**. In an embodiment,

optical signals are received from optical fiber **1190**, into planar waveguides **1165** formed from the inventive dielectric stack **1140** and routed to optoelectrical or optical device **1160** for processing, re-routing, or conversion to electrical signals, for example. In some embodiments, optoelectrical die **1160** are connected to one or more electrical devices **1162** and integrated electrical devices **1164** via metal lines **1126** in the interconnect layer **1120**. In embodiments, optical fiber **1190** is aligned to discrete waveguide **1165**, formed from inventive dielectric stack **1140**, which is further aligned to optical device **1160** to allow for the receiving and sending of optical signals for processing, re-routing, or conversion to electrical signals, for example. Optical alignment of devices to the waveguide, in embodiments, provides less than 1 dB power loss and in other embodiments, less than 0.5 dB. In preferred embodiments, power loss is much less than 0.5 dB. Accurate alignment between the optical fiber and the discrete waveguide **1165** fabricated from the inventive dielectric stack **1140**, and between the dielectric stack **1140** and the optical or optoelectrical device **1160**, is necessary to reduce potential power loss to tolerable levels.

(142) In some embodiments, the optical signals originate on the submount assembly **1105** and are transmitted through planar dielectric waveguide structure **1140** to the optical fiber **1190**. In yet other embodiments, the signals are received from the optical fiber **1190** for one or more of processing, routing, and conversion to electrical signals.

(143) Referring to FIG. **12A**, the steps of forming a dielectric interposer with a patterned waveguide from the inventive dielectric stack structure are shown that include a providing step **1000**, a depositing step **1210**, and a patterning step **1220**.

(144) In providing step **1200**, a substrate is provided with one or more optoelectrical or electrical devices coupled to an interconnection layer. In embodiments, these devices are one or more of a transistor, capacitor, resistor, inductor, or other electrical device, or an array of one or more electrical devices. In other embodiments, these devices are one or more of a p-channel metal oxide semiconductor (PMOS) transistor and an n-channel metal oxide semiconductor (NMOS) device or devices. In yet other embodiments, the devices are an array of transistors based on complementary metal oxide semiconductor (CMOS) transistors technology. In yet other embodiments, the one or more devices coupled to the interconnection layer as described in providing step **1200** in FIG. **12A** is a bipolar transistor, two or more bipolar transistors, or an array of bipolar transistor devices. In yet other embodiments, the one or more devices is a field effect transistor, two or more field effect transistors, or an array of field effect transistors. In some embodiments, transistor arrays coupled to the interconnect layer are used for signal processing, signal conditioning, signal generation, memory, and computation, for example.

(145) In depositing step **1210**, a stack of dielectric layers is deposited on the substrate to form the unpatterned inventive dielectric stack on the substrate, which is then patterned in patterning step **1220** to form the inventive interposer. In some embodiments, the patterned dielectric stack structure can be a section of waveguide aligned to an optical or electrical device, for example, for the transmission of optical signals to and from an optical fiber connected to the submount assembly. In other embodiments, these waveguides can include sections of the inventive dielectric stack that are patterned spot size converters, filters, arrayed waveguides, multiplexers, demultiplexers, gratings, power combiners, and the like. In yet other embodiments, these waveguides can provide part of a mechanical structure for the formation of hermetic seals. In yet other embodiments, these waveguides can be a combination of one or more of these types of structures fabricated from the inventive dielectric stack structure. In yet other embodiments, the buffer layer and the layers of the repeated stack are patterned to form a filter, an arrayed waveguide, a grating, a multiplexer, a demultiplexer, a spot size converter, or a power combiner, and the like.

(146) In embodiments, the patterning step **1220** is used to pattern the blanket dielectric stack structures into one or more planar waveguides. Patterning steps can include the use of established photoresist layers, used either directly as a mask for wet or dry etching or etch processing, or via a photoresist layer used to transfer a pattern from the photoresist to a hard mask which is utilized for

wet or dry etching or etch patterning of the inventive dielectric film stack. Processes for photoresist patterning and subsequent wet and dry etching of film structures are well established for those skilled in the art of dielectric film patterning techniques.

(147) Referring to FIG. **12B**, steps of forming a submount assembly with the inventive interposer are shown that include providing step **1240**, a first coupling step **1250**, and a second coupling step **1260**. In providing step **1240**, a substrate is provided wherein the substrate includes at least a first device coupled to an interconnection layer, wherein the substrate includes a waveguide patterned from a stack of dielectric layers. Patterned waveguide structures include filters, arrayed waveguides, gratings, multiplexers, demultiplexers, spot size converters, power combiners, and the like. In the first coupling step **1250**, a second device is coupled to the substrate, wherein the device is configured to interface between the waveguide and the at least a device. In embodiments, the second device is a receiving device, for example, such as a photodiode for receiving optical signals transmitted through the waveguide and subsequently converting the optical signals to electrical signals that are delivered to the interconnect layer. Conversely, in other embodiments, the second device is a sending device, for example, such as a laser for converting electrical signals from the interconnect layer, for example, to optical signals for transmission to the waveguide. In the second coupling step **1260**, an optical fiber is coupled to the substrate, wherein the optical fiber is configured to interface with the waveguide. Optical fibers are typically used in communication networks for the transmission of optical signals between submount assemblies and over long distances. By contrast, planar waveguides and the transmission of optical signals in free space are used to transmit optical signals within submount assemblies and over short distances. Optical fibers that are used to deliver optical signals are typically connected to the substrate and aligned with waveguides or other devices, such as a lens, to provide the necessary interface for transferring the optical signals from the fiber to the submount assembly to which the optical fiber is connected.

(148) Referring to FIG. **13A**, a perspective view of interposer **1300** is shown for an embodiment that includes inventive dielectric film stack **1340** patterned to form a waveguide, a v-groove **1392** for coupling and aligning an optical fiber to the interposer **1300**, and x-y-z stop structure **1366** for aligning devices to the patterned dielectric stack **1340**. In the embodiment shown in FIG. **13A**, x-y-z stop structure **1366** is a single element. In other embodiments, any one of the x-stop, y-stop, and z stop can be combined to facilitate the alignment of optical and optoelectrical devices to the submount assembly. In yet other embodiments, the x-stop, a y-stop, and a z-stop can be in one or more individual parts, or multiple parts, to provide the same function of aligning devices in each of the x, y, and z directions identified in FIG. **13A**. In other embodiments, one or more stops are provided for one of the x, y, and z directions. In yet other embodiments, one or more stops are provided for two or three of the x, y, and z directions. And in yet other embodiments, multiple stops are provided for one or more of the x, y, and z directions. In yet other embodiments, one or more alignment marks are provided in addition to the stops. In yet other embodiments, alignment marks are provided to align the optical, optoelectrical, and electrical devices without the stops.

(149) Referring to FIG. **13B**, a cross sectional schematic of an embodiment for submount assembly **1305** is shown that is formed on interconnect layer **1320** on substrate **1310** with inventive dielectric stack **1340**. In the embodiment shown in FIG. **13B**, features **1367**, **1368**, **1369** are provided for the alignment of optical or electrical device **1360** to the planar waveguide fabricated from the dielectric stack **1340**. Optoelectrical device **1360** is connected through buffer layer **1330** to metal layer **1326**. Metal layers **1326** are insulated with intermetal dielectric **1327** in interconnect layer **1320**. In some embodiments, interconnect metal layers **1326** connect optoelectrical devices **1360** to integrated electrical devices **1364** in the substrate **1310** or to other devices in the submount assembly **1305**. Alignment of optical/optoelectronic device **1360** is required to align the optical sending or receiving side **1361** of optical or optoelectrical device **1360** to the planar waveguide formed from the inventive dielectric stack structure **1340** and to thereby allow for the transfer of optical signals between the planar waveguide formed from the inventive dielectric stack structure **1340** and the

optical or optoelectrical device **1360**. It is important to note that for embodiments in which the device **1360** is an optical device, alignment is required within the submount assembly **1305** to provide for the transfer of optical signals between the planar waveguides and the device **1360** in the submount assembly **1305**, but not necessarily for electrical connections. In some embodiments, however, metal bond pads are implemented to attach optical devices **1360**. Alignment of the planar waveguides formed from the inventive dielectric stack structure **1340** to optical fiber **1390** is achieved in some preferred embodiments with v-groove **1392** in substrate **1310**.

(150) In an embodiment shown in FIG. **13B**, substrate **1310** is shown with optional integrated device **1364**. Integrated electrical devices **1364**, in preferred embodiments, are connected to the interconnect layer **1320**. Interconnect layer **1320** is a structure of metal lines **1326** and intermetal dielectric layers **1327** that provide insulated conductive pathways for interconnecting electrical and optoelectrical devices and dies that are fabricated on, mounted in, or connected from an external submount assembly to the submount assembly **1305**. In preferred embodiments in which optoelectrical die **1360** are mounted onto the submount assembly **1305**, metal interconnects **1326** are routed within the interconnect layer **1320** to provide electrical connections for electrical and optoelectrical devices in, on, or connected to the submount assembly **1305**, and to the underlying electrical devices **1364**.

(151) Alignment marks **1365** are provided in some preferred embodiments for the alignment of optical, electrical, and optoelectrical devices on the submount assembly **1305**. In some embodiments, alignment marks are provided in the buffer layer **1330** or the top layer of the interconnect layer **1320** of the interposer **1300** for alignment of devices, such as the optoelectrical device **1360**, within the submount assembly. Alternatively, alignment marks can be provided in other layers, on or in, the substrate. In preferred embodiments, alignment mark **1369** is for optical alignment, as is used in automated die placement tools for example, to position the die onto the submount assembly **1305**. Alignment mark **1369** in embodiments is a patterned feature in or on a layer or the substrate in the submount assembly **1305**. In some embodiments, the patterned features are an ink mark, a coloration mark, or discoloration mark of the top or another layer in the substrate or in one of the layers on the substrate. In some embodiments, the alignment mark is a means of providing optical contrast. Alignment mark **1369** in some embodiments is one or more of an etched feature, a deposited feature, a laser scribed feature, a feature created by exposure to an electron beam, or an ion milled feature.

(152) Alignment features **1367** and **1368** provide physical stops for the alignment of optical die **1360**, and other devices on the submount assembly **1305**. Accurate placement of devices and waveguides on optical submount assemblies is necessary for the transmission of the optical signals through the optical circuit on the submount assembly **1305**. In instances for which optical devices and features are not aligned, significant loss of the optical signal can occur, and in extreme circumstances can result in complete loss or blockage of the optical signal. Stop **1368**, in the embodiment shown in FIG. **13B**, is a z-direction stop, in that this stop is intended to fix the height (in the z-direction) of optoelectric device **1360** on the submount assembly **1305**. Stop **1367**, also in the embodiment shown in FIG. **13B**, is an x-direction stop, in that this stop is intended to fix the location of the optoelectric device **1360** in the x-direction as referenced in FIG. **13A** on the submount assembly **1305**. In some embodiments, a y-direction stop is also included. And in yet other embodiments, one or more of an x-direction stop, a y-direction stop, and a z-direction stop are provided. In yet other embodiments, one or more stops are provided for each device **1360** mounted on the submount assembly **1305** that requires alignment.

(153) Additionally, in preferred embodiments, a v-groove feature **1392** or other alignment feature is provided to align the optical fiber **1390** to the submount assembly **1305** and to planar waveguides formed from the inventive dielectric stack **1340**.

(154) Referring to FIG. **14**, steps for forming a dielectric interposer with a patterned waveguide from the inventive dielectric stack structure are shown that include a providing step **1400**, a first

forming step **1410**, a second forming step **1420**, and a third forming step **1430** as described herein. In providing step **1400**, a substrate is provided wherein the substrate includes an interconnection layer (see interconnect layer **1320**, for example.) In a first forming step **1410**, a waveguide is formed that includes the inventive dielectric film structure on the substrate. In a second forming step **1420**, at least one of an x-stop, a y-stop, a z-stop, and an alignment mark are formed on the substrate wherein the x-stop, a y-stop, a z-stop, and an alignment mark are configured to align a device with the waveguide. In a third forming step **1430**, at least one alignment feature is formed on the substrate wherein the alignment feature is configured to align an optical fiber with the waveguide.

(155) A specific benefit and feature of the planar dielectric waveguide structure is that in addition to its primary use for fabricating optical waveguides, it can also be used to produce mechanical features such as the alignment stops. In some embodiments, for example, the inventive dielectric stack is patterned using photoresist, for example, and then etched to a depth to establish the z-direction height, and for example, to create features for x-direction and y-direction stops as required. The capability to produce stops from the dielectric stack material, outside of the waveguide areas, provides an added benefit in implementing the planar dielectric stack structure on the inventive interposer. The use of the dielectric stack film stack to produce mechanical features such as the structures described for alignment stops and marks, as well other features described herein, is particularly enabled by the achievable thickness ranges of the inventive dielectric stacks. Thicknesses on the order of 2-25 micrometers are of the same thickness ranges that are suitable for alignment marks and stops. By combining the highly accurate vertical dimensioning capability that is achievable with highly controllable additive deposition technology with the highly controllable subtractive dry and wet etch technology, the relative heights of alignment features and stop features formed from the dielectric stack film structures can be formed with high accuracy. In addition to the applicable thickness benefits, the accuracy in the lateral dimensioning of the stops is generally provided by photolithographic patterning processes, which are highly accurate to within small fractions of a micrometer.

(156) Referring to FIG. **15A**, submount assembly **1505**, formed from interposer **1500**, is shown that includes substrate **1510**, interconnect layer **1520**, and inventive dielectric stack **1540**. Dielectric stack **1540** is patterned to form inventive planar dielectric waveguide. Interconnect layer **1520** is a structure of metal lines **1526** and intermetal dielectric **1527**. Metal lines **1526** provide electrically conductive pathways for interconnecting electrical and optoelectrical devices and dies that are fabricated on, mounted in, or connected from an external submount assembly to the dielectric interposer **1500**. In a preferred embodiment, one or more optoelectrical die **1560** are mounted onto the interposer **1500**, and the metal interconnects **1526** are routed within the interconnect layer **1520** to provide electrical connections for electrical and optoelectrical devices in, on, or connected to the interposer **1500**, or submount assembly **1505**, and to underlying integrated electrical devices in the substrate, if present. It is to be understood that optical devices can be mounted with metal bond pads **1522**, as means for mechanical bonding, without the specific requirement for electrical connections to other devices on the submount assembly **1505**. A discrete waveguide (see **1140**, for example) may not require electrical interconnection to other devices on the submount assembly **1505**, but the same or similar bonding methodologies that are used for to provide mechanical bonding and electrical interconnection can be utilized to bond the optical device **1560** to the submount assembly **1505**.

(157) In optical circuits, and in particular, in optical circuits within which lasers are utilized for converting electrical signals to optical signals, significant levels of heat can be generated that may require dissipation in some embodiments to prevent premature failure of, or damage to, a submount assembly or components mounted on the submount assembly. In addition to lasers, other optical, electrical, and optoelectrical devices can generate significant levels of heat while in operation. Submount assemblies, therefore, in some embodiments, would benefit from design features that

facilitate heat dissipation. In the inventive submount assembly **1505**, one or more of a thermally conductive dielectric layer is incorporated into the inventive submount assembly **1505** with the inventive dielectric stack **1540** to facilitate dissipation of thermal energy from the submount assembly **1505**.

(158) In the cross section shown in FIG. **15A** of an embodiment for the inventive submount assembly **1505**, a thermally conductive dielectric layer **1528** is disposed between the substrate **1510** and the interconnect layer **1520**. In these and other embodiments, the thermally conductive dielectric material, such as aluminum nitride, for example, is combined with inventive submount assembly **1505** in conjunction with heat generating optoelectrical devices **1560** and inventive planar dielectric stack **1540**. In embodiments, inclusion of heat-dissipating, thermally conductive dielectric layer **1528** with inventive dielectric stack structure **1540** improves the reliability of the submount assembly **1505** by providing thermally conductive pathways that allow for the transferring of heat from heat generating devices **1560** to heat sinks connected to the substrate **1510** or the submount assembly **1505**. In preferred embodiments, thermally conductive dielectric layer **1528** is aluminum nitride or an alloy of aluminum nitride. In other embodiments, other thermally conductive dielectric material is used in submount assembly **1505** in conjunction with the optoelectrical devices **1560** and inventive planar dielectric stack **1540**. In other embodiments, materials that are electrically conductive, such as the metal traces **1526** that are used in the interconnect layer **1520**, are used to transfer heat from heat generating devices **1560** to the thermally conductive layers **1528** for conduction of heat to heat sinks on the submount assembly **1505**.

(159) In other embodiments, as for example shown in FIG. **15B**, a thermally conductive dielectric layer **1529** is positioned within the interconnect layer **1520**. The metal traces **1526** in interconnect layer **1520**, which are commonly composed of aluminum, copper, other metal, or combination of metals, generally have a high thermal conductivity, and can provide heat dissipation pathways from the heat generating optoelectronic device **1560** to the thermally conductive dielectric material **1529**. The thermally conductive dielectric material **1529** is used in some embodiments to provide pathways that allow for the transferring of heat from the heat generating devices **1560** to one or more heat sinks connected to the submount assembly **1505**.

(160) Referring to FIG. **16A**, a sequence of steps for forming a substrate with a thermally conductive layer and an interconnection layer used in embodiments of the inventive dielectric interposer **1500** is shown. These steps, which include the formation of a thermally conductive layer are shown that include a providing step **1600**, a first forming step **1610**, and a second forming step **1620**, as described herein. In providing step **1600**, a substrate is provided whereon a thermally conductive layer is formed in first forming step **1610**. In the second forming step **1620**, an interconnection layer is formed on the thermally conductive layer. The sequence of steps shown in FIG. **16A** is one method for preparing a substrate with a thermal layer **1580** and an interconnect layer **1520** in preparation for the deposition of the inventive dielectric stack **1540**, and the resulting formation of the inventive dielectric interposer **1500**. In embodiments, the thermal layer **1528** is a heat sink for the removal of excess heat from devices **1560**, for example. In other embodiments, the thermal layer **1528** provides a thermally conductive pathway for the transfer of heat from heat-generating devices **1560** to heat sinks on or connected in some way to the sub-mount assembly **1505**.

(161) In FIG. **16B**, steps for forming other embodiments of the inventive dielectric interposer **1500** with a thermally conductive dielectric layer are shown. The steps in FIG. **16B** include a providing step **1640**, a first forming step **1650**, a second forming step **1660**, and a third forming step **1670**. In providing step **1640**, a substrate **1510** is provided for the dielectric interposer **1500**. In first forming step **1650**, a whole or part of an interconnect layer is formed on the substrate **1510**. In some embodiments, first forming step **1650** includes the formation of a part of the interconnect layer, that is, one or more layers of the interconnect layer **1520** but not the complete thickness of the

interconnect layer. In other embodiments the thermally conductive dielectric layer **1529** is formed on the interconnect layer **1520**. It is important to note that the thermally conductive layer can be formed at one or more of various positions in the interposer structure **1500** and remain within the scope of the current invention. Embodiments for the thermally conductive dielectric layer, for the purposes of providing a heat sink or a pathway to a heat sink include one or more of a thermally conductive layer **1528** on the substrate **1510**, a thermally conductive layer **1529** on the interconnect layer **1520**, and a thermally conductive layer **1529** within the interconnect layer **1520**, as described herein. The thermally conductive layer, in some embodiments, is partially at one height in the interconnect layer **1520**, and is partially at one or more other heights in the interconnect layer **1520**. For example, a thermally conductive layer **1528** may be on the substrate **1510** for part of the submount assembly **1505** and then partially at another height within the interconnect layer **1520**. In these embodiments, connections between the levels of the thermally conductive layers can be provided using the same thermally conductive material as in the thermally conductive layers **1529**, a metal layer **1526**, or an intermetal dielectric **1527**. In preferred embodiments, the use of the same thermally conductive material to connect multiple thermally conductive layers **1528**, **1529** is expected to produce the most efficient heat transfer although this approach might also have increased processing costs in some embodiments.

(162) The second forming step **1660**, for embodiments in which the thermally conductive layer **1529** is formed within the interconnect layer **1520**, is typically followed by completion of the remaining layers of the interconnect layer **1520**. In these embodiments, electrical connections **1526** may be required in some embodiments through the thermally conductive dielectric layer **1529** to connect underlying integrated electrical devices (see integrated device **964**, for example) or to connect underlying connection layers **1526** that reside below the thermally conductive layer **1529**. Third forming step **1670** includes the forming of an electrical connection in or through the dielectric layer that contains a thermally conductive dielectric layer **1529** to one or more of the interconnection layers **1526** that reside in the dielectric layer and in some embodiments to underlying integrated electrical devices (see integrated electrical device **964**, for example). Similarly, for embodiments in which the thermally conductive layer **1529** is deposited on the complete, or partially completed, interconnect layer **1520**, third forming step **1670** also includes the forming of electrical connections **1526** through the thermally conductive dielectric layer **1529** and the forming of one or more connections in or through this thermally conductive layer **1529** to one or more of the interconnection layers **1526** that reside below the thermally conductive dielectric layer **1529**. In embodiments in which the interconnect layer **1520** is nearly completed, the thermally conductive layer **1529** may form the uppermost dielectric layer in the structure of the interconnect layer **1520**.

(163) It is important to note that the thermally conductive layer **1528**, **1529** can be incorporated into the inventive interposer **1500** in various ways and remain within the scope of the current invention. FIG. **16B** shows steps in the formation of some embodiments of the inventive dielectric interposer **1500** with the inventive dielectric stack **1540** for which a thermally conductive dielectric layer **1528**, **1529** is included as described herein. In embodiments, the thermally conductive dielectric layer **1528**, **1529** is one or more of a heat sink for the removal of excess heat from devices **1560**, for example, and a thermally conductive pathway for the transfer of heat from heat-generating devices **1560** to heat sinks on or connected to the sub-mount assembly **1505**. The combination of heat-removing layers **1528**, **1529** with the heat generating devices **1560** and integrated planar dielectric waveguides formed from the inventive dielectric stack structure **1540** are beneficial for enhancing the reliability of submount assemblies that are uniquely enabled by this combination.

(164) Referring to FIG. **17A**, a cross sectional schematic of an unpatterned inventive dielectric stack **1740** is shown for embodiments of the inventive dielectric interposer **1700** formed on substrate **1710** for embodiments with optional thermally conductive layer **1728**, interconnect layer

1720, and buffer layer **1730**. Patterning of inventive dielectric stack **1740** from FIG. **17A** yields inventive dielectric stack section **1740a** and inventive dielectric stack section **1740b** as shown in the cross-sectional schematic in FIG. **17B**. In preferred embodiments, dielectric stack section **1740a** and dielectric stack section **1740b** form cavity **1794**. The inset in FIG. **17B** shows a perspective view of the top surface of inventive interposer **1700** after patterning of the dielectric stack **1740** to form dielectric stack sections **1740a**, **1740b**, and the cavity **1746**.

(165) Referring to FIG. **17C**, a schematic cross section of inventive dielectric interposer **1705** is shown with optoelectrical device **1760** within cavity **1794**. In some embodiments, the optoelectrical device **1760** is connected with bond pad **1722** to the underlying metallization **1726** in the interconnect layer **1720** through openings in buffer layer **1730**. Metallization traces **1726** generally form interconnections between the various electrical devices on and within the interposer and are shown for general demonstrative purposes in FIG. **17**, and in other figures, and not intended to show a specific patterns or structures for the interconnections. Metallization layers **1726** provide interconnections between electrical and optoelectrical devices mounted onto the interposer **1700**, to integrated electrical devices in the substrate (see integrated electrical device **964**, for example), and to other devices and other submount assemblies connected to submount assembly **1705**.

(166) In FIG. **17D**, a cross sectional schematic of cap **1796** on submount assembly **1705** to create capped optoelectronic package **1708** is shown. In some embodiments, cap **1796** is provided to seal the cavity **1794**, and to provide hermetically sealed protection of the submount assembly within the cavity **1794**. Cap **1796** is coupled to the cavity walls formed from dielectric stack sections **1740a**, **1740b**, formed from the inventive dielectric stack structure **1740**, to cover and protect the optoelectric devices **1760** mounted within the cavity **1746**. In typical preferred embodiments, a metal seal **1797** is utilized to bond the cap **1796** to the cavity walls **1740a**, **1740b**. In other embodiments, the seal or bond layer **1797** between the cap **1796** and the mechanical supports can be made from materials such as adhesive resins, solder material, and the like. The cap **1796** is shown mounted directly on the inventive dielectric stack structure **1740**, but it should be understood that additional layers can be formed above the dielectric stack structure **1740** for various reasons that include one or more of improved bonding layer adhesion, vertical height adjustment, alignment, and provision for positional stops, among other reasons, and remain within the scope of the current invention.

(167) In FIG. **18**, the steps for providing a capped submount assembly **1708** from inventive submount assembly **1705** using inventive dielectric stack structure **1740** are shown and include a providing step **1800**, a first forming step **1810**, a first patterning step **1820**, a second forming step **1830**, and a third forming step **1840** as described herein for some embodiments. In the providing step **1800**, a substrate **1710** that includes an interconnection layer **1720** is provided. In some embodiments, the substrate **1710** has a thermally conductive layer **1728** on substrate **1710** or within the interconnect layer **1720**. In some other embodiments, the substrate **1710** does not have a thermally conductive layer **1728** on or in substrate **1710**, or on or within the interconnect layer **1720**. In a first forming step **1810**, inventive dielectric stack **1740** is deposited onto the interconnect layer **1720**. Inventive dielectric stack **1740** is patterned in first patterning step **1820** to form a waveguide from the inventive dielectric stack **1740** and one or more support structures **1740a** and **1740b** that are also formed from the inventive dielectric stack **1740**. An embodiment of support structures is shown for example in FIGS. **17B-D**. In a second forming step **1830**, a device **1760**, for example, is formed on the substrate, wherein the device is configured to couple to the waveguide formed from the dielectric stack **1740**. In the third forming step **1840**, a cap **1796** is positioned to cover the device that is coupled to the waveguide, by disposing the cap **1796** on the dielectric stack structure **1740** patterned to form a waveguide **1740a**, which also serves as a mechanical support structure, and the support structures **1740b**. The benefit of using the inventive dielectric stack as both a mechanical support and a waveguide enables the use of the waveguide to transmit light

signals into the cavity and out from devices mounted within the cavity while providing a capability for hermetic sealing. The transmission of light through planar waveguides formed from the inventive dielectric film structures **1740** can be used to facilitate the transmission or receiving, or both, of optical signals from optical fibers mounted external to the cavity, through the cavity walls **1740a**, to or from devices **1760** mounted within the cavity **1794**.

(168) In the cross sections of the embodiments for the inventive interposers and submount assemblies shown and described herein, it should be understood that waveguides fabricated from inventive dielectric stack **140**, **740**, **940**, **1040**, and **1140**, in some embodiments can be a small section of waveguide aligned to an optical or electrical device, for example, for the transmission of optical signals to and from an optical fiber connected to the submount assembly. In other embodiments, these waveguides can include sections of the inventive dielectric stack **140** that are patterned spot size converters, filters, arrayed waveguides, multiplexers, demultiplexers, gratings, power combiners, and the like. In yet other embodiments, these waveguides can provide part of a mechanical structure for the formation of hermetic seals. In yet other embodiments, these waveguides can be a combination of one or more of these types of structures fabricated from the inventive dielectric stack structure **140**. In yet other embodiments, the buffer layer and the layers of the repeated stack are patterned to form a filter, an arrayed waveguide, a grating, a multiplexer, a demultiplexer, a spot size converter, or a power combiner.

(169) The foregoing disclosure of the exemplary embodiments of the present invention has been presented for purposes of illustration and description. It is not intended to be exhaustive or to limit the invention to the precise forms disclosed. Many variations and modifications of the embodiments described herein will be apparent to one of ordinary skill in the art in light of the above disclosure. The scope of the invention is to be defined only by the claims appended hereto, and by their equivalents.

(170) Further, in describing representative embodiments of the present invention, the specification may have presented the method and/or process of the present invention as a particular sequence of steps. However, to the extent that the method or process does not rely on the particular order of steps set forth herein, the method or process should not be limited to the particular sequence of steps described. As one of ordinary skill in the art would appreciate, other sequences of steps may be possible. Therefore, the particular order of the steps set forth in the specification should not be construed as limitations on the claims. In addition, the claims directed to the method and/or process of the present invention should not be limited to the performance of their steps in the order written, and one skilled in the art can readily appreciate that the sequences may be varied and remain within the spirit and scope of the present invention.

Claims

1. A structure comprising a substrate, wherein the substrate comprises an interconnection layer comprising an electrical interconnection line, a waveguide disposed on the interconnection layer, wherein the waveguide comprises a stack of multiple layers of silicon oxynitride films, wherein the waveguide comprises a core comprising silicon oxynitride, wherein at least a silicon oxynitride film in the multiple layers of silicon oxynitride films is deposited using a plasma enhanced chemical vapor deposition (PECVD) process, wherein process gases in the PECVD process comprise a silicon-containing precursor not containing hydrogen and one or more precursors for oxygen and nitrogen, and wherein the one or more precursors of oxygen and nitrogen comprise an amount of hydrogen less than 10% of the process gases.
2. A structure as in claim 1, wherein the amount of hydrogen is less than 5% of the process gases.
3. A structure as in claim 1, wherein the process gases comprise a hydrogen-free chemistry.
4. A structure as in claim 1, wherein the process gases comprises less than 10% ammonia.
5. A structure as in claim 1, wherein the one or more precursors for oxygen and nitrogen comprise

at least one of a precursor for oxygen and a precursor for nitrogen, a precursor for oxygen, and a precursor for nitrogen and oxygen, or a precursor for nitrogen, and a precursor for nitrogen and oxygen.

6. A structure as in claim 1, wherein the multiple layers of silicon oxynitride films comprise an index of refraction between 1.55 and 2.05 and a stress level less than 25 MPa in magnitude, achieved by configuring ratios of process gases.

7. A structure as in claim 1, wherein the at least a silicon oxynitride film comprises a stress level less than 25 MPa in magnitude, or wherein the stack of silicon oxynitride films comprises a stress level less than 25 MPa in magnitude, or wherein individual silicon oxynitride films in the stack of silicon oxynitride films have a stress level less than 25 MPa in magnitude.

8. A structure as in claim 1, wherein the at least a silicon oxynitride film is formed using process temperatures less than 500 C to prevent damages to the substrate.

9. A structure as in claim 1, the interconnection layer is formed on an electrical device, and wherein the interconnection layer comprises an electrical interconnection line electrically connected to a terminal of the electrical device.

10. A structure comprising a substrate, wherein the substrate comprises an interconnection layer comprising an electrical interconnection line, a waveguide disposed on the interconnection layer, wherein the waveguide comprises a stack of multiple layers of silicon oxynitride films, wherein the waveguide comprises a core comprising silicon oxynitride, wherein at least a silicon oxynitride film in the multiple layers of silicon oxynitride films is deposited using a plasma enhanced chemical vapor deposition (PECVD) process, and wherein process gases in the PECVD process comprise an amount of hydrogen configured to form the at least a silicon oxynitride film having a stress level less than 25 MPa in magnitude or configured to form the at least a silicon oxynitride film using process temperatures less than 500 C to prevent damages to the substrate.

11. A structure as in claim 10, wherein the process gases comprises less than 10% ammonia.

12. A structure as in claim 10, wherein the amount of hydrogen is less than 10% of the process gases.

13. A structure as in claim 10, wherein the process gases comprises a precursor for silicon, a precursor for nitrogen, and a precursor for oxygen, and wherein at least one of the precursor for silicon, the oxygen precursor, or the nitrogen precursor comprises a hydrogen-free chemistry.

14. A structure as in claim 10, wherein the at least a silicon oxynitride film is formed using process temperatures less than 400 C to prevent damages to the substrate.

15. A structure as in claim 10, wherein the process gases comprise a hydrogen-free chemistry.

16. A structure comprising a waveguide disposed on a substrate, wherein the substrate comprises an interconnection layer formed on an electrical device, and wherein the interconnection layer comprises an electrical interconnection line electrically connected to a terminal of the electrical device, wherein the waveguide comprises a stack of multiple layers of silicon oxynitride films, wherein the waveguide comprises a core comprising silicon oxynitride, wherein at least a silicon oxynitride film in the multiple layers of silicon oxynitride films is deposited using a plasma enhanced chemical vapor deposition (PECVD) process, and wherein process gases in the PECVD process comprise an amount of hydrogen configured to form the at least a silicon oxynitride film, as-deposited, having a level of optical loss of less than 1 dB/cm.

17. A structure as in claim 16, wherein the amount of hydrogen is less than 10% of the process gases.

18. A structure as in claim 16, wherein process gases comprise a silicon-containing precursor not containing hydrogen.

19. A structure as in claim 16, wherein the waveguide is formed using process temperatures less than 500 C to prevent damages to the substrate, and wherein ratios of process gases are configured to achieve a stress level less than 25 MPa in magnitude.

20. A structure as in claim 16, wherein the process gases comprises less than 10% ammonia.

